fMRI hyperscanning reveals widespread neural coupling during natural conversations

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Abstract

Conversation is one of the most powerful tools that people have for building and maintaining social connections. How does conversation bring two minds together? Here we tracked the neural dynamics between dyads during real-time conversation using fMRI hyperscanning, where both people were scanned simultaneously, to explore whether and how conversation aligns brains. Dyads engaged in natural, free-flowing conversation as they co-constructed the content of their dialogue. In a control condition, partners read aloud scripted dialogue from another dyad's conversation, preserving the turn-taking structure while removing the co-construction. To examine neural alignment, we compared whole-brain intersubject correlation between partners in each condition, assessing both synchronous and predictive neural dynamics. Results showed that naturalistic conversation led to significant widespread neural alignment across brain regions implicated in attention, affect, executive control, and social cognition, compared to scripted conversation. Temporally refined analyses showed that during naturalistic conversation, relative to scripted conversation, both speakers and listeners anticipated each other within a short temporal window, with brain activity preceding their partner's in the same neural systems. Conversation may help people communicate by aligning neural activity. This work lays the foundation for future research into how neural alignment allows people to create shared understanding and form social bonds.

Significance statement

This study investigated how conversation aligns the brains of interacting individuals. Scanning pairs of people's brains simultaneously, we found that spontaneous conversations led to widespread neural alignment, particularly in networks associated with higher-order processes, such as attention, affect, executive control, and social cognition. Listeners anticipated speakers and speakers anticipated listeners activity within a narrow timeframe in these networks, evidence that naturalistic interaction involves proactive engagement by both the speaker and the listener. These findings highlight the active role of predictive processing in spontaneous dialogue, offering new insights into the neural dynamics underlying verbal two-way communication.

Introduction

Conversation is one of the most powerful tools that people have for building and maintaining social connections. In everyday life, people engage in conversation frequently—sharing stories with friends, working through problems with colleagues, or making small talk with strangers. These interactions allow for the exchange of information and carry the potential to create shared understanding (Nguyen et al., 2022). We know that conversation facilitates this exchange, but how exactly does conversation bring two minds together in service of communication?

When people converse, their thoughts, interpretations, and emotions intertwine in a dynamic dance, each mind responding to the cues of the other. As conversation partners interact, alignment cascades across multiple levels. Conversation partners often exhibit physiological alignment, including aligned heart rates (Feldman et al., 2011; Pérez et al., 2021), skin conductance, and bodily movements, such as gestures and postures (Palumbo et al., 2017; Prochazkova et al., 2021; Reed et al., 2013). Partners also coordinate their eye gaze patterns as they establish joint attention to regulate the flow of interaction (Wohltjen & Wheatley, 2024). These forms of physiological alignment may reflect deeper social connections, as they often mirror joint emotional arousal and shared engagement (Pérez et al., 2021).

Conversations also align people's language. Partners often adapt their language to each other by using similar words, matching syntactic structures, and mimicking speech rhythms, tone, and prosody (Garrod & Pickering, 2004; Yeomans et al., 2022). Linguistic alignment may enhance predictability and mutual comprehension, allowing partners to respond more quickly and effectively to each other (Templeton et al., 2022). That is, linguistic alignment reflects more than the words people say, but also the automatic and intentional mirroring of communicative behaviors that maintain the flow of conversation and foster mutual understanding.

Physiological and linguistic alignments may serve as a foundation for more abstract, psychological alignment. Through conversation, individuals tune to each other's mental states as they create shared meaning (Stolk et al., 2014), shared experience (Gorman et al., 2013), and a shared worldview (J. Chen et al., 2017a; Zadbood et al., 2017). This tuning can impact deep, meaningful representations, by aligning each individuals' thoughts, beliefs (Pickering & Garrod, 2006; Sievers et al., 2024), and self-perceptions (Welker et al., 2024). Alignment at this level does not imply that individuals think the same way or agree on everything (Speer et al., 2024b). Rather, it reflects processes where mental convergence and divergence can foster a shared understanding, within a particular context.

Although we know that conversation aligns people's bodies, words, and minds, we do not know *how* that alignment arises. Physiological and linguistic alignment alone cannot explain the full extent of psychological alignment. Neuroimaging during asynchronous communication, where participants passively listen to recorded narratives, reveals that listeners' neural activity mirrors speakers'

activity, particularly in regions linked to language comprehension and social cognition (Silbert et al., 2014; Stephens et al., 2010). This neural coupling captures a meaningful indicator of communication success (Schoot et al., 2016), such that greater synchrony is associated with greater comprehension (Stephens et al., 2010), learning (Nguyen et al., 2022), and shared goals and emotional states (Anders et al., 2020). This research on asynchronous communication demonstrates the potential for measures of neural synchrony to tap into psychological alignment. However, the magic of real-time conversation, with its dynamic reciprocity between speaker and listener, cannot be fully captured by asynchronous paradigms that study each component in isolation.

Research on synchronous, naturalistic communication has used simultaneous neuroimaging (i.e., hyperscanning) to further demonstrate the value of neural synchrony for social interaction (Tsoi et al., 2022). For example, EEG and fNIRS hyperscanning studies of two-person, bidirectional communication have likewise shown that speakers and listeners exhibit inter-brain synchronization (Pérez et al., 2017), that the magnitude of the observed coupling increases when speaker and listener engage in turn-taking dialogue (Jiang et al., 2012; Liu et al., 2017), and that synchrony is greater in romantic couples than strangers (Kinreich et al., 2017). Hyperscanning can also provide further insight into the cognitive and social processes by which conversation aligns minds, beyond studies of asynchronous communication (Hasson et al., 2012; Schoot et al., 2016). In particular, hyperscanning expands our ability to test unique hypotheses about predictive processing in conversation offered by asynchronous studies (e.g., Dikker et al., 2014). In particular, we expect to see that communicators' brains not only mirror, but anticipate each others' activity in real time.

Though prior work has established that both synchronous and asynchronous communication aligns brains, two key gaps remain. First, it is unclear in this prior work if neural alignment captures simply time-locked production and comprehension of language during conversation, or also the higher-level shared meaning-making that emerges when partners co-create content in real time (Hamilton, 2021). Understanding how alignment arises during conversation would require isolating the active dialogue component of conversation. Most hyperscanning research lack a control that preserves the turn-taking structure while removing the co-construction component. Here, we do so by comparing scripted turn-taking, where utterances are fixed in advance to free-flowing conversation, where partners generate content in real-time. We predict that these dynamic, co-created conversations, where partners must adapt moment-to-moment, responding directly to each other's words, expressions, and intentions, will align brains significantly more than during passive, interleaved speaking and listening. That said, the simplified attention context and shared stimuli offered by scripted conversation may enhance synchrony. Alternatively, passive and active communication may rely on the same underlying neural mechanisms.

Second, most prior work examining neural alignment during active, dynamic, real-time, turn-taking conversations, have primarily used EEG, MEG, ECoG, or

fNIRS, which are limited to surface-level cortical activity (Dikker et al., 2017; Jiang et al., 2012; Schwartz et al., 2024; Zada et al., 2024). However, studies of asynchronous communication highlight the value of whole-brain signals, including alignment in medial and subcortical areas crucial for social cognition and emotional processing regions that can only be captured by fMRI (Nummenmaa et al., 2014; Silbert et al., 2014; Stephens et al., 2010; Zadbood et al., 2017). Here we use fMRI hyperscanning to overcome this limitation, allowing us to observe neural activity in high spatial resolution throughout the entire brain (Misaki et al., 2020). Moreover, fMRI hyperscanning allows us to examine distributed activity patterns rather than just localized activation levels, which can help identify how whole systems engage during interaction (Tsoi et al., 2022). This approach enables us to pinpoint the cognitive processes engaged during social interactions, identify moments when these processes are active, and explore how partners' experiences relate to each other.

In the present study, we leverage fMRI hyperscanning to study neural alignment associated with interpersonal mental engagement during naturalistic conversations, over and above the process of simple turn-taking. We capture time-locked synchrony as our primary measure of neural alignment. In addition, we capture lagged synchrony to capture how speakers' and/or listeners' brains anticipate one another during dynamic conversation. Dyads in this study engaged in two types of interactions: free-flowing, spontaneous conversations and scripted conversations (Figure 1). By comparing these two conditions, we test the hypothesis that spontaneous, real-time conversation leads to greater neural synchrony and prediction relative to similarly-structured, but scripted conversational exchange. This tight control allows us to disentangle the unique contributions of real-time alignment from those of a predetermined dialogue and to characterize the brain regions and networks that support this alignment. Our study focuses on the real-time dynamics of spontaneous conversations, providing a new lens for investigating the neural mechanisms of social interaction.

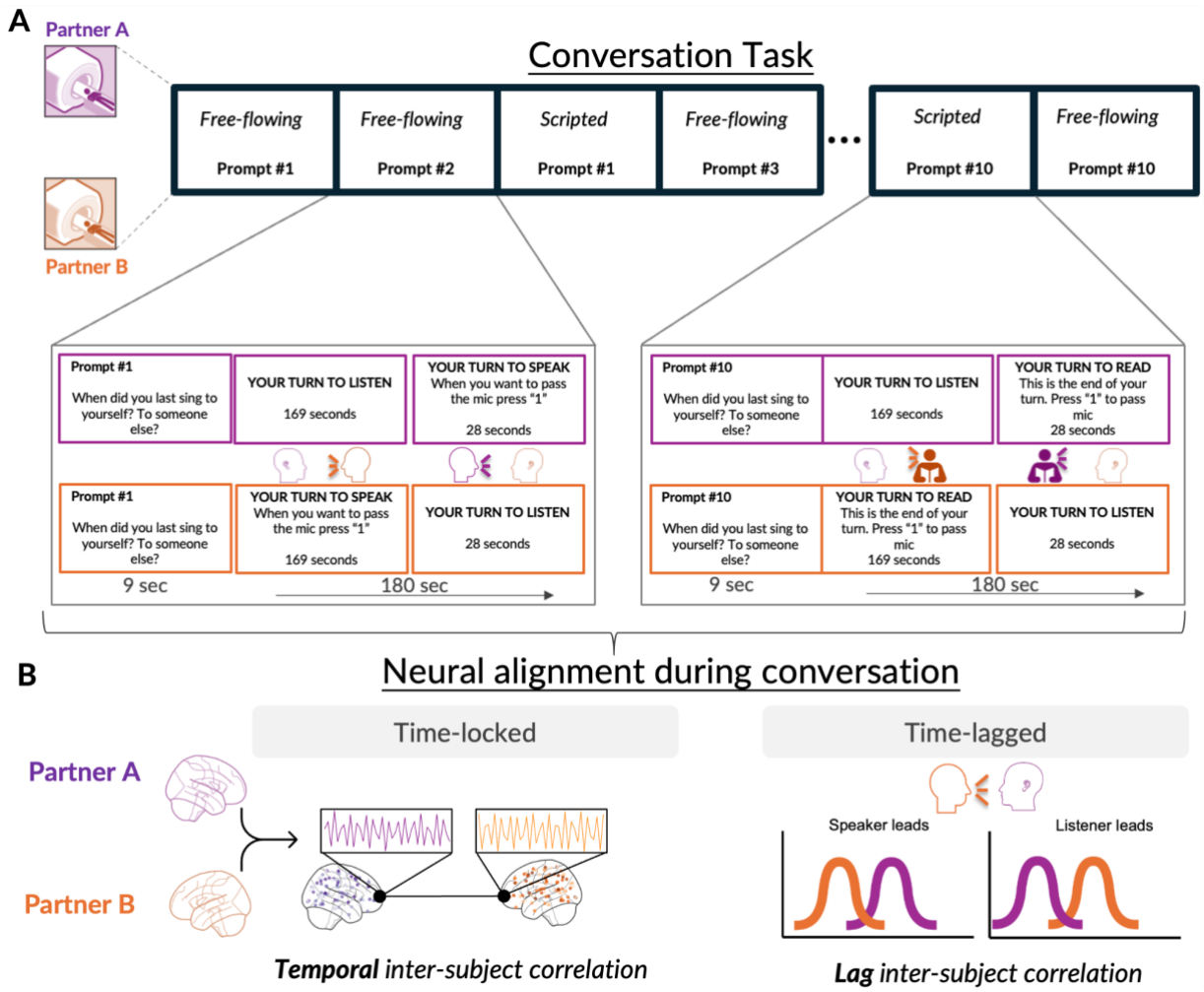

Figure 1. Overview of the analytic framework for capturing neural alignment during conversation. (A) Participants were scanned in pairs while performing the conversation task. One each free-flowing trial, they were shown a discussion prompt for 9 seconds, and then one participant spoke while the other listened. Speakers pressed a button to pass the speaker role to the listener. (B) We measured intersubject correlation of the dyad's neural activity within each parcel using two approaches: (1) time-locked: similarity of concurrent timecourses of brain activity; and (2) time-lagged: similarity of both the timecourse of brain activity, shifted in time to see how the speaker predicts the listener, and vice versa.

Results

To investigate how conversation aligns neural activity between two individuals, we measured temporal inter-subject correlation (ISC). Temporal ISC quantifies the degree to which the overall timecourse of brain activity in each neural region is synchronized within a dyad. We measured ISC within each turn (Figure 1A) to capture both synchronous (time-locked) and predictive (time-lagged) neural dynamics between speaker and listener during conversation. Time-locked ISC was measured at lag 0 (Figure 1B), where the neural activity of the conversation partners is fully synchronous. Time-lagged ISC reflects the listener's brain activity either following (lags < 0) or preceding (lags > 0) that of the speaker's, thus providing insight into predictive activity between partners. This approach enables us to capture both concurrent and predictive neural alignment that characterizes the dynamic interplay between conversation partners.

These ISC analyses were applied to two conversation conditions: free-flowing and scripted. In the free-flowing condition, dyads were provided with an initial prompt to spark discussion but were free to take the conversation in any direction, allowing it to flow naturally and spontaneously in real-time. In the scripted condition, dyads read a scripted dialogue from another dyad's conversation, ensuring a parallel, controlled turn-taking conversational structure, but without any need for generating personal content or actively listening in order to respond. By comparing these conditions, we aimed to discern how free-flowing and structured dialogue differentially elicit neural alignment. Analyses were conducted at both the region level (within specific brain parcels) and the network level (across brain systems).

Conversation evokes widespread neural alignment across the brain

To formally test for time-locked neural alignment during the conversation, we examined temporal ISC in both conditions within the the following seven intrinsic networks defined by Yeo et al. (2011): visual, somatomotor, dorsal attention, ventral attention, limbic, frontoparietal, and default (Figure 2D, see Table S3 for statistical results). Across both conditions, we see widespread neural alignment. A linear mixed effects model revealed a main effect of network ($F(6, 344161) = 396.65$, $p < 0.001$), indicating that some networks were overall more aligned than others (see Table S6 for pairwise comparison between all networks).

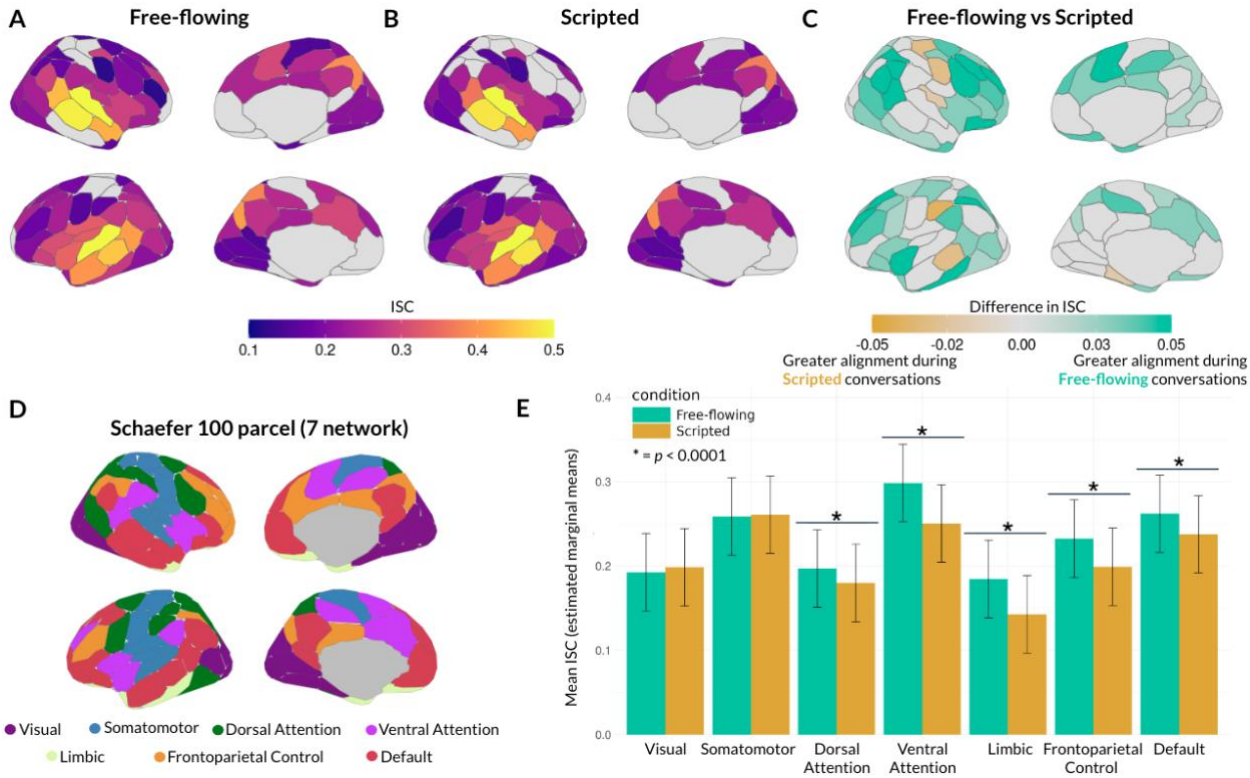

Figure 2. Neural alignment during conversation. Brain activity within a dyad was significantly correlated across the cortex in the (A) free-flowing and (B) scripted conditions. (C) The free-flowing conversations aligned neural activity to a greater extent than the scripted conversations in many regions throughout the brain (green), though we observed exceptions– regions where scripted conversations elicited greater alignment (yellow). All results are corrected for false-discovery rate (FDR) and thresholded at $q < 0.05$. (D) The atlas used to parcellate the cortex with each region and measure neural alignment. The colors of each parcel represents a neural network assignment (defined by Yeo et al., 2011). (E) Average ISC across dyads within neural networks for the free-flowing (green) and scripted (yellow) conditions. The error bars represent standard error. *$p < 0.0001$.

Looking first within the free-flowing condition (Figure 2A), we see that alignment is most prominent in the ventral attention network ($r = 0.30$), including in regions associated with salience detection (right temporal parietal junction (TPJ), anterior mid-cingulate cortex, and anterior insula (AI)). We also find strong alignment in the default network ($r = 0.26$), including in regions implicated in language comprehension (inferior frontal gyrus, middle frontal gyrus; superior frontal gyrus), areas involved in semantic processes (e.g., inferior parietal lobule), and self-referential processes (e.g., posterior cingulate cortex). We also see high neural alignment in the somatomotor network ($r = 0.26$), particularly in bilateral primary auditory cortex; the frontoparietal control network ($r = 0.23$) implicated in executive functioning, particularly in the right dorsolateral prefrontal cortex; the limbic network ($r = 0.18$) involved in affective processing, particularly the bilateral cingulate cortex); and the visual network ($r = 0.19$), particularly bilateral visual cortex.

This broad engagement demonstrates that real-time conversation aligns multiple higher-order functional processes in both partners during free-flowing conversation. Psychologically, this could occur because of the content and spontaneity of the conversation, or merely the shared stimuli and structure of turn-taking inherent in any dialogue. Indeed, in the scripted condition (Table S6), dyads were also aligned across all networks and showed a similar pattern of alignment across the cortex as the free-flowing condition Figure 2B. Alignment was particularly high in the somatomotor (r = 0.26), ventral attention(r = 0.25), and default (r = 0.24) networks.

Although both free-flowing and scripted conditions lead to widespread neural alignment (Figure 2A-B), a linear mixed effects model also revealed a main effect of condition ($F(1, 344147) = 33.47$, $p < 0.001$), such that the free-flowing conversation evoked significantly more alignment than scripted conversations. In addition, a significant interaction between network and condition ($F(6, 344143) = 28.32$, $p < 0.001$; Figure 2C,E) revealed the specific networks, where alignment differed significantly during free-flowing versus scripted conversations (Table S4). When compared to the scripted condition, neural alignment was significantly greater during free-flowing conversation in the dorsal attention network ($\beta = 0.02$, SE = 0.004, $p < 0.0001$), particularly in the superior parietal lobule; the ventral attention network ($\beta = 0.04$, SE = 0.005, $p < 0.0001$) particularly in bilateral AI and vlPFC; the limbic network ($\beta = 0.04$, SE = 0.007, $p < 0.0001$), particularly in the orbitofrontal cortex and temporal pole; the frontoparietal control network ($\beta = 0.03$, SE = 0.004, $p < 0.0001$) in the dlPFC; and in the default network ($\beta = 0.02$, SE = 0.003, $p < 0.0001$), particularly in right TPJ, inferior temporal gyrus, and vlPFC.

Scripted conversations did not elicit significantly greater alignment than free flowing conversations in any networks (Table S5). Scripted conversations did show significantly greater alignment in specific regions associated with motor, affect, memory, and language processing, likely due to reading scripted dialogue. These included bilateral primary motor cortex, right posterior insula, and left parahippocampal gyrus and middle temporal gyrus. Overall, these findings demonstrate that real-time, naturalistic conversation leads to more extensive and distributed neural alignment across higher-order process brain regions compared to structured, scripted dialogue.

Speaker and listener alignment across temporal lags

Free-flowing conversation evokes widespread time-locked neural alignment across the brain. However, time-locked alignment is not the only neural dynamic relevant to successful communication. During naturalistic conversation, speakers might consider how their words will land with their listeners, and listeners anticipate what the speaker may say next. To capture how partners predict each other over the conversation, we measured ISC at multiple temporal offsets (lags) between speaker and listener during each turn. For example, at a lag of -2 TR (-3 seconds), we

assessed the extent to which the speaker anticipates the listener by correlating the timecourse of the speaker's neural activity during each speaking turn with the listener's neural timecourse of the same length starting -3 seconds after the onset of the turn. If the speaker were to anticipate the listener, we would expect that the peak of the correlation would occur at -3 seconds. If instead the peak of the correlation occurs at -1 TRs (-1.5 seconds) then this would indicate that the speaker is not anticipating the listener that far ahead of time. By considering both negative lags (in which the *speaker's* neural response precedes the listener's) and positive lags (in which the *listener's* neural response precedes the speaker's), we aimed to characterize the interplay of integrative and predictive processing during naturalistic conversation. In particular, we used this analysis to test for larger correlations at shorter offsets, and any asymmetry between the magnitude of speaker and listener predictive activity.

Speakers anticipate listeners

During free-flowing conversations, speakers anticipated the listener during a narrow temporal window when compared to scripted conversation in higher-order process networks. Particularly at earlier temporal offsets (-1.5 seconds), free-flowing conversations elicited greater alignment across all higher-order process networks compared to scripted conversations (Figure 3C). This effect was most pronounced in the limbic and frontoparietal networks, systems implicated in affective processing and executive control, as well as within the dorsal and ventral attention networks and default network. Speakers' anticipatory activity remained sustained (up until +4.5 seconds) specifically in the limbic and frontoparietal control. Specific regions showing greater anticipatory alignment included bilateral orbitofrontal cortex (OFC), temporal pole, AI, ITG, AI, dlPFC, and in right superior parietal lobule at earlier offsets (-1.5 seconds; Figure S2C). These findings suggest that when the conversation is free-flowing, speakers anticipate the listener by engaging affective and control processes that immediately precede the listener's neural activity.

At longer negative offsets (-4.5 to -6 seconds), speakers show little anticipatory activity in both scripted and free-flowing conversations. If anything, speakers' anticipatory activity is negatively correlated with listeners activity in the free-flowing conversations, such that a contrast between the two conditions reveals significantly reduced anticipatory alignment by the speaker during free-flowing relative to scripted conversations across all networks (Figure 3C, Figure S2C). This reversal is strongest in the visual and dorsal attention networks, particularly in regions such as primary somatomotor, V1, and left STG and TPJ. These findings highlight that spontaneous versus structured communication contexts amplify the extent to which speakers anticipate the listener's neural activity, particularly at short but not longer lags.

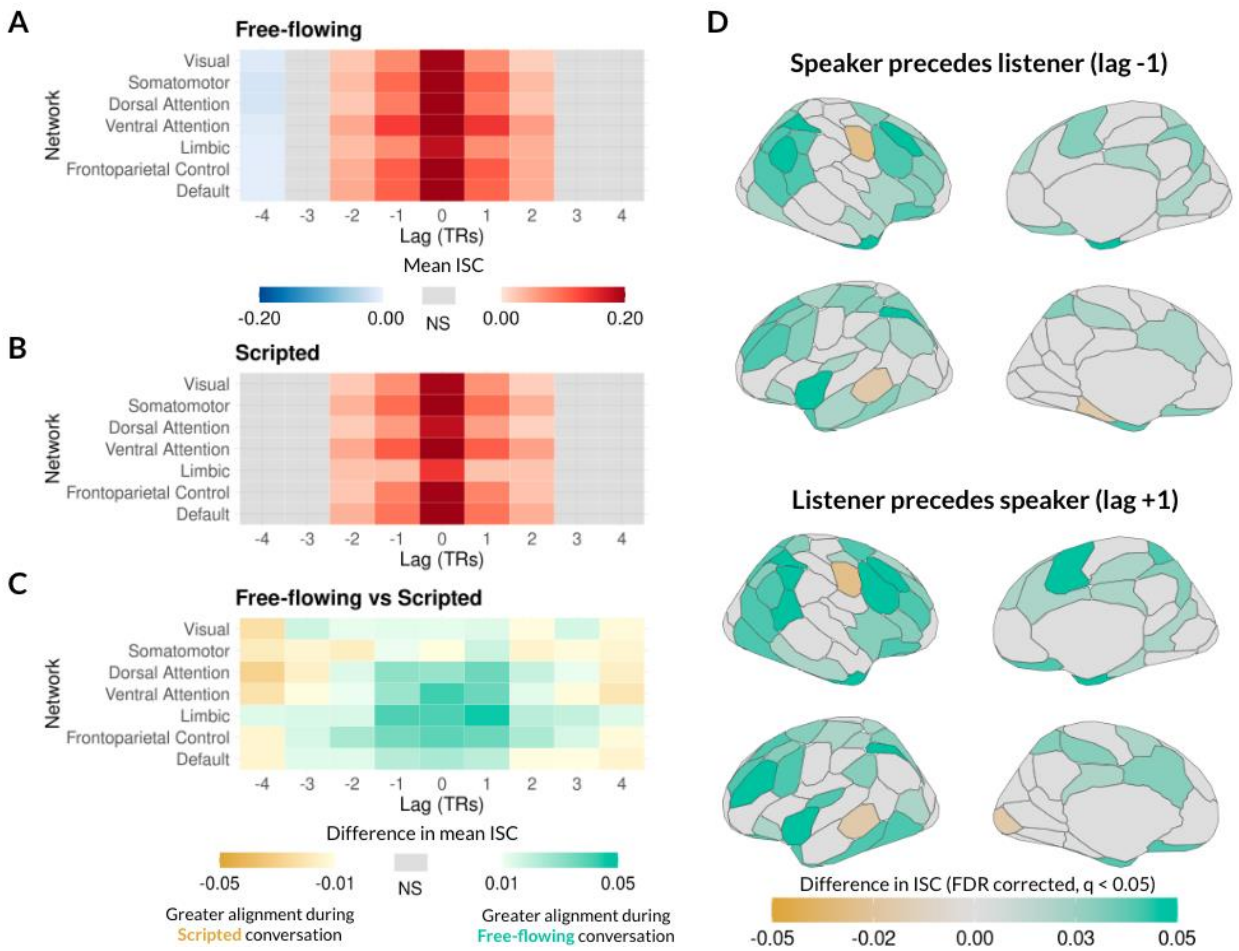

Figure 3. Temporal lagged ISC results. ISC values at each parcel between speaker and listener at varying temporal lags for the (A) free-flowing (B) scripted, and (C) free-flowing > scripted. The listener's data is shifted -4 to +4 TRs (±6 seconds) before calculating temporal ISC. Lag 0 reflects time-locked ISC. Greater ISC indicates greater alignment between speaker and listener. For the condition contrast, green indicates greater alignment for the free-flowing condition, and yellow greater alignment for the scripted condition. (D) The alignment pattern of the free-flowing > scripted contrast is shown on the brain surface for lag -1 (top) and lag +1 (bottom). Lag -2 (-1.5 seconds) reflects brain regions where the speaker's activity precedes the listener's more significantly during the free-flowing condition. Lag +2 (+1.5 seconds) patterns of alignment where the listener's activity precedes that of the speaker during the free-flowing relative to the scripted condition. All results shown are FDR corrected at $q < 0.05$.

Listeners anticipate speakers

During free-flowing conversations, listeners also anticipated the speaker's neural activity at short, but not long lags relative to scripted conversations. By examining positive lags, where the listener's neural activity preceded that of the speaker, we see that tree-flowing conversations showed greater anticipatory alignment particularly at short offsets (+1.5 seconds) across dorsal attention, ventral attention, limbic, frontoparietal control and default networks (Figure 3C). As with negative lags, we also find that listeners show greater alignment specifically in the

limbic and frontoparietal control networks that remained sustained (up until +4.5 seconds), and in the dorsal attention network (up to +3 seconds). The limbic network which is involved in affective processes and the frontoparietal network which is implicated in responding in executive control. Listeners showed consistently greater predictive alignment across multiple positive lags in regions across the frontal cortex, anterior insula, bilateral somatomotor cortex, bilateral ITG, right temporal pole, right precuneus, left mPFC during free-flowing compared to scripted conversations (Figure S2F). Together, these findings point to stronger predictive processing by listeners in areas related to self-referential, mentalizing, and affective processing during spontaneous dialogue.

At longer temporal offsets (+6 seconds), overall alignment was low in both conditions (Figure S2D-E). Indeed, at these longer lags, we again see that listeners become anticorrelated with speakers in the free-flowing condition, leading to significantly lower predictive alignment than in the scripted condition in the dorsal attention, ventral attention, and default networks for listeners (Figure 3C). This reversal is strongest in bilateral dlPFC, superior parietal lobule, cingulate gyrus, and left ITG and IPL/TPJ at later offsets (+6 seconds; Figure S2D).

Overall, these comparisons between conditions indicate that naturalistic conversations drive listeners to actively engage and predict speakers' activity, at short offsets and across all higher-order neural systems. These results parallel those we see for speakers' anticipatory activity, which also ramp up within a narrow temporal window. These symmetrical anticipatory processes may help speakers and listeners both rise to the unique demands of spontaneous communication by crafting production with listeners in mind, and responding adaptively, in real-time.

Discussion

Conversation offers a fundamental way for people to connect. This study investigated how conversation brings two minds together by examining neural alignment during real-time conversation between dyads. Using fMRI hyperscanning, we found that spontaneous, free-flowing conversations led to widespread neural alignment across multiple higher-order process networks associated with attention, affect, executive control, and social cognition, compared to scripted conversations. Time-lagged analysis further revealed that listeners anticipated speakers and that speakers anticipated listeners to a greater extent during natural, active dialogue than scripted dialogue, particularly within a narrow temporal offset. Together these results reveal compelling evidence of whether, when, and how neural synchrony manifests as minds come together during dynamic, natural conversation.

Past research has shown that shared experiences, such as watching a movie or listening to a story, lead to neural synchrony in regions that support auditory and visual processing, language production and comprehension (e.g., STS, TPJ; Stephens et al., 2010; Hasson et al., 2012) and higher-order process systems like the default network (Yeshurun et al., 2017). These findings from unidirectional communication have been critical in expanding our understanding of how our brains support our ability to understand and communicate with each other through language. Here, we show that during bidirectional, real-time conversation, neural alignment emerges not only in regions implicated in prior studies of narrative comprehension and production, but also across multiple higher-order process systems important for supporting the attentional and cognitive demands of active, spontaneous conversations, and shared meaning making.

The current study was designed to consider the unique role of co-construction of conversation in neural alignment relative to other forms of shared experience. Free-flowing conversations indeed led to significantly more cortical alignment across all higher-order neural networks relative to scripted conversations. The observed alignment across multiple networks supports theories of verbal communication born out of studies on asynchronous communication that emphasize the alignment of neurocognitive systems (Hasson et al., 2012; Yeshurun et al., 2017). Even asynchronous communication can synchronize distributed networks, and in doing so facilitate shared language processing, emotional processing, perspective-taking, and attentional focus. Our results underscore the power of spontaneous, dynamic, interactive nature of spontaneous conversations to align minds over and above mere stimulus entrainment. In doing so, we emphasize the potential for conversation to foster social connection by aligning distributed brain systems, which will be important for laying the groundwork for future studies examining the neural mechanisms that underlie meaningful communication and social connection.

Although prior work has emphasized sensory and language networks, our results reveal that compared to scripted conversations, free-flowing conversations elicited alignment in multiple systems. Namely, we also found greater alignment

across the dorsal and ventral attention, limbic, frontoparietal control, and default networks during free-flowing conversation. This suggests that naturalistic dialogue involves networks supporting both bottom-up and top-down attention, affective processing, executive control, and mentalizing, likely because of the additional demands of co-creating conversation content absent in scripted dialogue. In free-flowing conversations, speakers must consider how the listener may hear their contributions, while the listener must carefully track what the speaker is saying in order to prepare a response. This shared need to select and prioritize information may help explain greater widespread alignment in attention and executive control systems. These findings provide new insight into how language is distributed across brains during natural conversation and echoes proposed theories that multiple networks are indeed involved in supporting human verbal communication (Fedorenko et al., 2024).

In addition to examining time-locked alignment during naturalistic conversation, we also explored how alignment unfolds over time for the speaker and listener as they actively generated and processed the content of their conversation. Though alignment was greatest at no lag, we found robust evidence of predictive neural activity in both the speaker and listener. Speaker-listener temporal dynamics in prior work report that the listener's neural response lagged behind that of the speaker at longer lags (up to -3 seconds; Stephen et al., 2010; Silbert et al., 2014). We also observed anticipatory speaker activity up to -3 seconds. We see robust, widespread anticipatory activity by the speaker, particularly in the free-flowing condition in the ventral attention, dorsal attention, limbic, frontoparietal control, and default networks. Listeners too play an active role in the conversation, showing nearly symmetrical anticipatory activity as they predict where the speaker might go next. Like speakers, we find that the listener's neural anticipation in free-flowing conversations, specifically, ramps up at short lags, especially in the dorsal attention, limbic and frontoparietal control networks, and becomes limited or anticorrelated at longer lags. Listeners are not passively responding to speaker input, but actively preparing for what will be said next. At longer lags, we see little, or even anticorrelated activity between speakers and listeners, suggesting that the turn-taking and real-time co-construction of dialogue may limit how far in advance the speaker is able to anticipate the listener (see Figure S3 for descriptive report of turn taken across trials). These results align with predictive coding frameworks and prior work on mentalizing during communication (Liu et al., 2020; Zadbood et al., 2017), and extend those findings to fully interactive, unscripted dialogue.

This study contributes to the growing body of work using hyperscanning to explore neural alignment during social interaction (Burns et al., 2025; Speer et al., 2024a; Speer, et al., 2024b; Zada et al., 2025). We see that conversational alignment emerges even in the absence of explicit coordination instructions, suggesting that natural conversation itself can function as a natural context for dynamic neural alignment. This aligns with recent findings showing that both consensus-building are marked by strong neural alignment (Sievers et al., 2024).

While methodologically distinct, these findings converge on the idea that neural alignment is a core feature of conversation.

That said, prior work showed that dyads who *diverged* rather than converged in mental state space during conversation had better conversations, both in terms of feelings of closeness to each other and agreement on policy issues (Speer et al., 2024a; Speer et al., 2024b). There are three differences between the current study and that conducted by Speer and colleagues (2024a), which together help reconcile these findings. First, we compared free-flowing vs scripted conversations whereas Speer and colleagues (2024a) contrasted conversation between friends and strangers. Neural alignment during free-flowing conversation may be a necessary precondition for divergence and exploration to emerge. Once dyads establish common ground, they can make larger conceptual leaps. In line with this, Sened and colleagues (2025) found that during longer conversations strangers first align until they reach common ground and then diverge, suggesting that neural alignment might be a stepping stone towards exploratory conversation and stronger social connection. Second, Speer and colleagues (2024a) incorporated self-reported conversation outcomes, linking neural dynamics to participants' subjective experience, whereas our ISC analysis focuses on temporal coordination rather than representational content, whereas our comparison of free-flowing and scripted conversation isolates structural differences in conversation. Lastly, the analytic approaches differ. ISC captures how much partners' brains co-fluctuate in time and is sensitive to the temporal coordination that supports mutual understanding, while mental-state–decoding framework used by Speer and colleagues (2024a) targets the *content* of cognitive and affective processes. Together, these approaches show that conversation can involve tightly coupled neural dynamics even when partners mentally construe the same events in different ways.

Although EEG and fNIRS studies have identified neural synchrony during communication (Bevilacqua et al., 2019; Jiang et al., 2012), fMRI allows for a higher spatial resolution, identifying specific regions and networks involved in conversation. This spatial precision adds to existing research on ISC by demonstrating how different brain regions and systems coordinate during real-time communication. For example, here we identify where two brains align both at the region-level and at the network-level. In addition [something about the multivariate results, if we are including those in the supplement]. The high spatial resolution offered by fMRI hyperscanning could be particularly valuable when integrated with linguistic encoding models to develop more detailed neural models of how information is transferred between speakers and listeners in real-time (Zada et al., 2025).

While these findings advance our understanding of real-time neural alignment, a few limitations should be acknowledged. First, our study focuses on short 3-min conversations, which may have limited our ability to capture more gradual forms of neural alignment that develop over extended interactions. Longer conversations are more enjoyable (Kardas et al., 2022) and can build friendships even among strangers (Aron et al., 1997). Future research examining how longer conversations affect both regional and network alignment could potentially offer a richer

understanding of how neural synchrony supports sustained social interaction (Speer et al., 2024b; Stolk et al., 2016). Second, the limited temporal resolution of fMRI restricts our ability to capture more rapid shifts in neural synchrony. Our results, even at low temporal resolution, highlight the need to examine alignment at multiple temporal scales. Doing so may be particularly important especially when examining conversation partners that respond rapidly to each other, such as those who feel a stronger sense of social connection to one another (Templeton et al., 2022). Here we show that the neural dynamics of naturalistic conversations: the speaker and listeners are not just neurally attuned to each other, but anticipate one another. Third, it may also be beneficial to take an integrative approach to examine different forms of alignment, especially considering that mental alignment is not the only route by which conversation partners navigate conversation (Speer et al., 2024a; Speer et al., 2024b). Fourth, while we see robust anticipatory activity by both speaker and listener, it is possible that autocorrelation from a slow hemodynamic response may confound these findings. That said, a pure autocorrelation account would be expected to affect conditions similarly and appear broadly across networks and lags. Instead, we see selective condition dependent differences that peak at different lags, which would argue against autocorrelation alone driving the result. Finally, while our analysis focused on ISC, using intersubject functional connectivity could provide deeper insights into how neural signals are coordinated across regions and networks during conversation (Regev et al., 2019). This approach could be useful in revealing how different brain systems contribute to the shared processing of conversational content, advancing our understanding of the neural mechanisms underlying social communication.

Conversations are central to human social connection, requiring both rapid comprehension and proactive engagement. This study provides compelling evidence that brains align during natural, free-flowing conversation, even more so than during other forms of shared experience, such as following a script. Natural conversation engaged a broad array of higher order neural systems, producing both timelocked intersubject correlation, as well as time-lagged anticipatory activation for both the speaker and listener in neural systems that would support the ability to prepare for and flexibly attend to ever-evolving conversational demands. Taken together, these findings reveal how real-time conversation brings minds together through the alignment of distributed brain systems, laying the groundwork for understanding how neural synchronization may facilitate social connection.

Materials and Methods

In this study, we investigate how real-time conversation impacts neural alignment between individuals using fMRI hyperscanning. We recruited dyads to participate in naturalistic and scripted conversations while their brain activity was simultaneously recorded from adjacent rooms. To enhance the accuracy of neural alignment measures, we functionally align participants' data using a narrative listening task (see Supplementary methods). We then used these alignment parameters to analyze brain activity during the conversation task, focusing on temporal inter-subject correlation (ISC). Our analyses examine the effects of conversation on neural synchrony, as well as the predictive relationship between speaker and listener brain activity. This approach allowed us to examine both fine-grained (parcel-level) and broader (network-level) patterns of neural alignment during conversation.

Participants

Sixty-three dyads (126 participants) engaged in a real-time conversation while undergoing fMRI hyperscanning. We excluded six dyads where one or both partner(s) had incomplete data due to technical difficulties during the conversation or hyperalignment tasks. The final dataset consisted of 57 dyads (n = 114 participants; M_{age}= 20.6,SD_{age} = 3.0; 73 female, 41 male, 0 non-binary; Asian = 40, Black and/or African American = 10, White = 34, Other or more than one race = 30). Thirty of the dyads were self-identified as friends and 27 were strangers who were randomly paired with one another, unacquainted before the study. All participants had to be at least 18 years old to be eligible for the study (Supplementary Table 1). All study procedures were reviewed and approved by the Institutional Review Board at Princeton University.

Conversation task

On each trial, dyads engaged in a 3-minute conversation that began with a specific prompt selected from the "Fast Friends" procedure, designed to foster social connection (Aron et al., 1997; Supplementary Table 2). Prompts progressively increased in intimacy, from low ("Would you like to be famous? In what way?"), to medium ("What do you value most in a friendship?"), to high ("Share with your partner an embarrassing moment in your life.").

Participants engaged in both naturalistic conversation (free-flowing condition), as well as a fully controlled conversation (scripted condition). In the free-flowing condition, dyads freely navigated and discussed topics: participants created their own responses to prompts, enabling spontaneous conversation, expression, and interaction. In the scripted condition, participants read from a pre-written script of a conversation from another pair. The scripted condition maintained a conversation's structure and turn-taking but without any personal content generation. Participants were randomly assigned whether to speak first for each prompt. The speaker

pressed a button to signal the end of their turn, prompting the other participant to speak.

Each dyad completed 20 trials, with condition (free-flowing vs. scripted) randomly assigned to each trial. Each run included two free-flowing and two scripted conversation blocks, beginning with a nine-second prompt and condition cue, followed by 180 seconds of conversation turn-taking. Participants were encouraged to use the full duration. Each run began and ended with a 12-second fixation cross. The entire protocol consisted of five runs, each lasting 13.6 minutes (544 TRs). This design, illustrated in Figure 1, allowed us to examine neural alignment during both spontaneous and pre-scripted conversations, providing insights into whether and how individuals align during conversation.

Imaging procedure

MRI data acquisition

The fMRI images for the dyads were collected using a 3T Siemens Skyra MRI system and a 3T Siemens Prisma MRI system. The same scanning parameters were used for both scanners. Functional scans were acquired with whole brain coverage in interleaved order (3.0 mm slice thickness, 3.0 × 3.0 mm in-plane resolution, flip angle = 80°). TE was 28 ms, and TR was 1500 ms. A T1-weighted image was acquired for anatomical reference (1.0 × .0 × 1.0 mm resolution, 176 sagittal slices, flip angle = 9°, TE = 2.98 ms, TR = 2300 ms). To minimize head movement, subjects' heads were stabilized with foam padding.

Imaging preprocessing

The fMRI data were preprocessed using fMRIPrep version 20.2.0, a Nipype-based tool chosen for its robustness and reproducibility in preprocessing diverse datasets. fMRIPrep automatically adapts workflows based on best-in-class algorithms, enabling high-quality preprocessing with minimal manual intervention. Detailed methods for preprocessing can be found in (Speer et al., 2024a).

Each T1-weighted (T1w) volume was corrected for intensity nonuniformity and skull stripped. Spatial normalization was performed using the International Consortium for Brain Mapping (ICBM) 152 Nonlinear Asymmetrical template version 2009c through nonlinear registration. Brain tissue segmentation of cerebrospinal fluid (CSF), white matter (WM), and gray matter was carried out on the brain-extracted T1w. Field map distortion correction involved coregistering the functional image to the same-subject T1w image, followed by boundary-based registration to the T1w with 9 degrees of freedom. Motion correction, field distortion correction, and spatial normalization were applied in a single step using Lanczos interpolation. Physiological noise regressors were extracted using CompCor, with six components each for temporal (tCompCor) and anatomical (aCompCor) variants.

To account for speaking-induced motion artifacts and distortions in the magnetic field, an additional denoising step was applied post-preprocessing. This involved performing confound regression on the preprocessed data using six head motion regressors, their cosine, first derivatives, squares, the white matter signal and its derivatives, the cerebrospinal fluid signal and its derivatives, and additional components as detailed in Speer et al. (2024a). This protocol was implemented after comparing 21 different confound regression models, and selecting the one that optimized ISC detection, before any hypothesis testing took place. In addition, seven task related regressors were included. This involved 4 boxcar regressors that were included to indicate the following: (1) the two trial conditions, (2) the speaking turns, (3) the listening turns, (4) and conversation prompt and three indicator regressors were initialized with zeros and filled with ones when (1) the participant pushed the button press, (2) their partner pressed the button, and (3) the screen changed. The residuals from this regression, representing the cleaned data, were then used for further analysis. These regressors were convolved with the HRF to account for the hemodynamics. After detrending, regressing the variables, and z-scoring the cleaned timeseries data were segmented into individual prompts (trials) to contrast the free-flowing and scripted conditions and to investigate trial effects over time.

Functional alignment

Traditional alignment methods in fMRI studies typically focus on anatomical features using a group anatomical atlas (Jenkinson et al., 2002), but there is significant heterogeneity across individuals in structure-function coupling (Haxby et al., 2001) . This variability can reduce the accuracy of comparing neural activity across subjects when using purely anatomical alignment methods (Bazeille et al., 2021). To address this limitation, we used functional alignment which aligns on functional response patterns rather than anatomical atlas. We specifically performed the functional alignment using the Shared Response Model (SRM) (Chen et al., 2014; Chen et al., 2015). To preserve the shared experience of each conversation within the dyad, while accounting for uniqueness across dyads. For each dyad, we used fastSRM (Kumar et al., 2021; Richard et al., 2019) to learn shared feature spaces for the dyad instead of learning a shared space for all participants. Twenty latent features are learned for each parcel, and once fit, the learned mappings are used to reconstruct the dyad's conversation data. We chose to functionally align at the dyad-level because the conversation trails of each dyad are unique to them, therefore, mapping their conversation data to the learned shared spaces of the dyad allows for functional alignment that more closely retains the fidelity of the individual characteristics of each dyad.

We functionally aligned dyads' brains by first defining dyad-specific hyperalignment parameters on independent data collected during a listening task that participants completed after the conversation. We applied the transformation parameters for each parcel to functionally align each of the 5 runs of the *conversation task*. This allowed us to project each parcel's voxel responses during the conversation into a common functional space, and enhance the functional

correspondence between partners (see Figure S1 for comparison with anatomical alignment).

Neural parcellation

We conducted all analyses at two levels. First, we extracted the time series data across 100-parcels, within the 7-network atlas (Schaefer et al., 2018). The parcellation approach, compared to conventional voxelwise and searchlight, is less computationally demanding for measuring neural alignment. Since the parcels are nonoverlapping, this approach eliminates the possibility of generating neuroanatomically implausible regions (Jolly & Chang, 2021). This approach facilitates inferences about both local and network-level effects. To assess network-specific model performance, we calculated the average model performance in the 7 resting-state functional networks (i.e., intrinsic networks) as defined by Yeo and colleagues (Yeo et al., 2011).

Statistical analyses

To examine whether and how conversations (free-flowing versus scripted) influence neural alignment between individuals, we conducted the following main analyses: First, we calculated inter-subject correlation (ISC) to examine dyadic neural alignment. We operationalized ISC by temporally aligned brain activity. Next, we examine how speaker activity predicts listener activity, by measuring timed-lagged speaker-listener temporal ISC. For each of these analyses, we looked at the main effects of conditions (free-flowing and scripted conversation) and their interaction of the two conditions (free-flowing > scripted). We performed these analyses at two scales: at the parcel level, focusing on localized brain regions, and at the network level, considering broader, functionally connected networks. This approach allowed us to assess both fine-grained and large-scale patterns of neural alignment, providing a better understanding of how conversations align brains across different contexts and scales.

Dyadic inter-subject correlation (Temporal ISC)

Temporal ISC focuses on correlations that measure alignment in neural activity across time within a dyad, capturing the degree to which the timing of brain activity is synchronized between participants during a shared experience. This approach assesses how similarly the brain activity of two individuals unfolds over time as they converse. Using the functionally aligned conversation data, we calculated the correlation between the neural activity of each dyad over time, within each trial. Temporal ISC measures the correlation between time series data of neural activity from two participants. For each *i* parcel, the correlation $r_{A,B}$ between the timeseries of partner A (ts_A) and B(ts_B) was calculated as:

$$r_{A,B} = cor(parcel_i_ts_A,\ parcel_i_ts_B)$$

For each of the 100 cortical parcels, we investigated whether dyads demonstrated significant neural alignment over the course of the conversation and whether this alignment differed between experimental conditions. Prior to modeling all ISC values were Fisher r-to-z transformed to ensure the correlation values followed a normal distribution before further analyses.

To identify parcels where neural alignment is statistically significant across all dyads over the course of the conversation, we used linear mixed-effects (LME) models with crossed-random effects (J. Chen et al., 2017b) using LME4 in R (Kuznetsova et al., 2017). This approach accounts for the nonindependence structure embedded in our data due to the repeated ISC measures across trials. For each of the 100 parcels, we fit a LME model with z-transformed ISC values as the dependent variable. Trial number and condition (free-flowing vs. scripted) were included as fixed effects, and a random intercept was included for each dyad to capture individual dyad-level variability.

After fitting the LME model for each parcel, the emmeans R package was used to obtain estimated marginal means for each condition. We then conducted two sets of tests: (1) the main effects test to identify parcels where ISC was significantly different from zero under each individual condition, and (2) pairwise contrast to identify parcels where ISC was greater during the free-flowing conversation when compared to the scripted conversation. All p-values were then adjusted for multiple comparisons across parcels using a false discovery rate corrected (FDR) *p*-values for multiple comparison at $q < 0.05$. Results from the model are plotted on the brain surface using the ggseg and ggsegSchaefer R packages (Mowinckle & Vadal-Piñeiro, 2020). This integrated approach allowed us to identify which parcels exhibited robust neural alignment and to determine whether that alignment differed significantly depending on the conversation conditions.

To characterize neural alignment at the network level and examine differences between conversation conditions, we conducted a LME model. Each of the 100 parcels was assigned to one of seven predefined neural networks based on the Schaefer et al., 2018 atlas. We fitted a single LME model to assess the impact of conversation conditions on ISC across all networks simultaneously. The model included condition (free-flowing vs. scripted), network, and their interaction (condition × network) as fixed effects. To account for individual variability and the nonindependence of observations within dyads, we included a random intercept for each dyad.

Following model fitting, we computed estimated marginal means for each combination of condition and network. We then performed two sets of statistical tests. First, we examined whether ISC within each condition (free-flowing and scripted) significantly differed from zero across all networks. Then, we evaluated whether ISC was significantly greater during free-flowing conversations compared to scripted conversations within each network. To account for multiple comparisons, all p-values were FDR corrected at $q < 0.05$.

Lagged synchrony/predictive activity

While a person is speaking, they need to anticipate how their words will land with their listener; while a person is a listener, they need to anticipate what the speaker will say next. We examined how speakers and listeners anticipate each other during naturalistic conversation using timed-lagged ISC across discrete conversation turns. First, the fMRI time series data were segmented into individual turns, each defined by a period during which one participant was the speaker and their partner was the listener. Although longer epochs have been recommended for reliable ISC estimation (Nastase et al., 2019), such durations may not always be feasible in natural conversational settings. To balance ecological validity with the need for stable correlation estimates, we excluded turns shorter than 14 TRs, thus removing segments most prone to spurious estimates while retaining sufficient data to capture meaningful temporal neural alignment dynamics.

Within the retained turns, we computed temporal ISC between the speaker's and listener's neural activity at varying temporal offsets (± 4 TRs or ± 6 seconds).These lags were chosen to probe anticipatory neural processes, reflecting how well one participant's brain activity might predict the other's upcoming activity. A positive lag correlation indicates that the listener's brain activity precedes and predicts the speaker's future activity, while a negative lag correlation indicates that the speaker's brain activity anticipates the listener's future responses.

For each lag and each parcel, we fit a LME model with z-transformed ISC as the dependent variable. Turn duration, trial number, and condition (e.g., free-flowing vs. scripted) were included as fixed effects, and a random intercept was included for each dyad to account for the nested structure of the data. In this way, we separately modeled and tested how ISC patterns varied with conversational dynamics across different temporal lags and conditions. The results were FDR corrected at $q < 0.05$.

Acknowledgments

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Code and data availability

Unless otherwise specified, all post-processing neuroimaging analyses were performed using Brainiak package in python. All code available to conduct these analyses will be made available here: (https://github.com/Princeton-Social-Neuroscience-Lab/PSNL-repo).

CRediT authorship contribution statement

Conceptualization: L.T., L.M.T., S.P.H.S., S.M.B., E.B.F., D.I.T.
Methodology: L.M.T., L.T., S.P.H.S., S.M.B., Z.Z., E.B.F., D.I.T.
Investigation: L.T., S.M.B.
Formal analysis for this paper: L.M.T., S.P.H.S.
Visualization: L.M.T.
Funding acquisition: L.T., E.B.F., D.I.T.
Project administration: E.B.F., D.I.T.
Supervision: E.B.F., D.I.T.
Writing – Original Draft: L.M.T.
Writing – Review and Editing: L.M.T., L.T., S.P.H.S., Z.Z., E.B.F., D.I.T.

Supplementary Materials

Supplementary methods

Narrative listening task

The first task was a narrative listening task, during which dyads were instructed to listen and pay attention to a story. Our motivation for including this data was to utilize it to functionally align the neuroimaging data for our main analyses. The stimuli used for the narrative task was an audio recording of 'I Knew You Were Black' told by Carol Daniel during a live taping of The Moth (audio freely available at https://themoth.org/stories/i-knew-you-were-black). The scan began with a 12-second instruction cue, followed by 13.3 minutes (534 TRs) of the 'I Knew You Were Black' audio while a fixation cross is presented in the middle of the screen, and a 12-second instruction cue that the experiment ended. The total length of the narrative listening task was 550 TRs (824 seconds).

Supplementary tables and figures

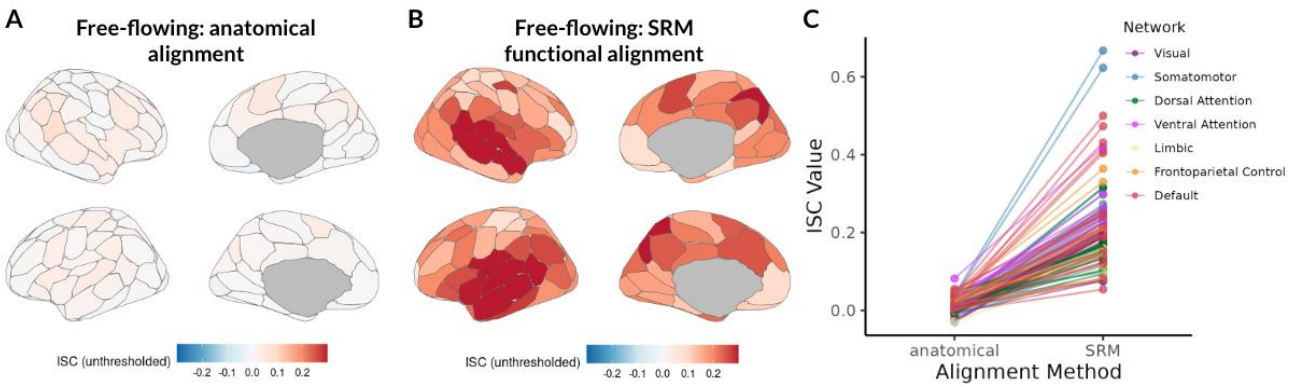
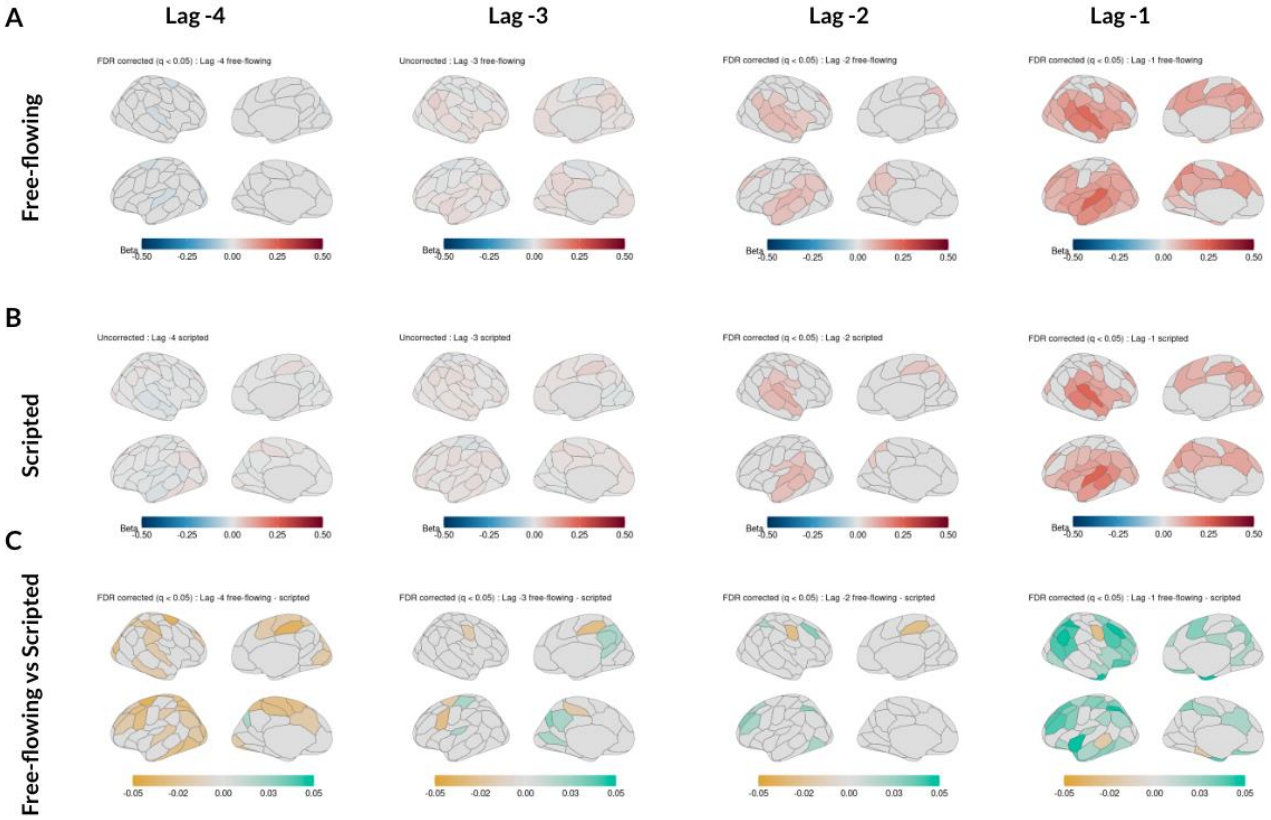


Fig S1. Shared Response Model (SRM) functional alignment enhances ISC during conversation. Unthresholded ISC maps during the free-flowing condition across the full sample (n = 57 dyads) after aligning the data using an (A) anatomical alignment approach and (B) after using SRM to functionally align the data. Prior to SRM alignment, the maximum ISC was 0.08 in the right middle temporal gyrus. After SRM alignment the maximum average ISC in the left primary auditory was 0.67. (C) Paired plot illustrating the difference in ISC using anatomical and SRM alignment approaches. Each parcel is colored according to the Yeo 7 network assignment. Wilcoxon signed-rank test shows that ISC significantly increased when using SRM alignment compared to anatomically alignment ($V = 5050$, $p < 0.001$), indicating an overall improvement in ISC following SRM alignment.

Speaker precedes listener



Listener precedes speaker

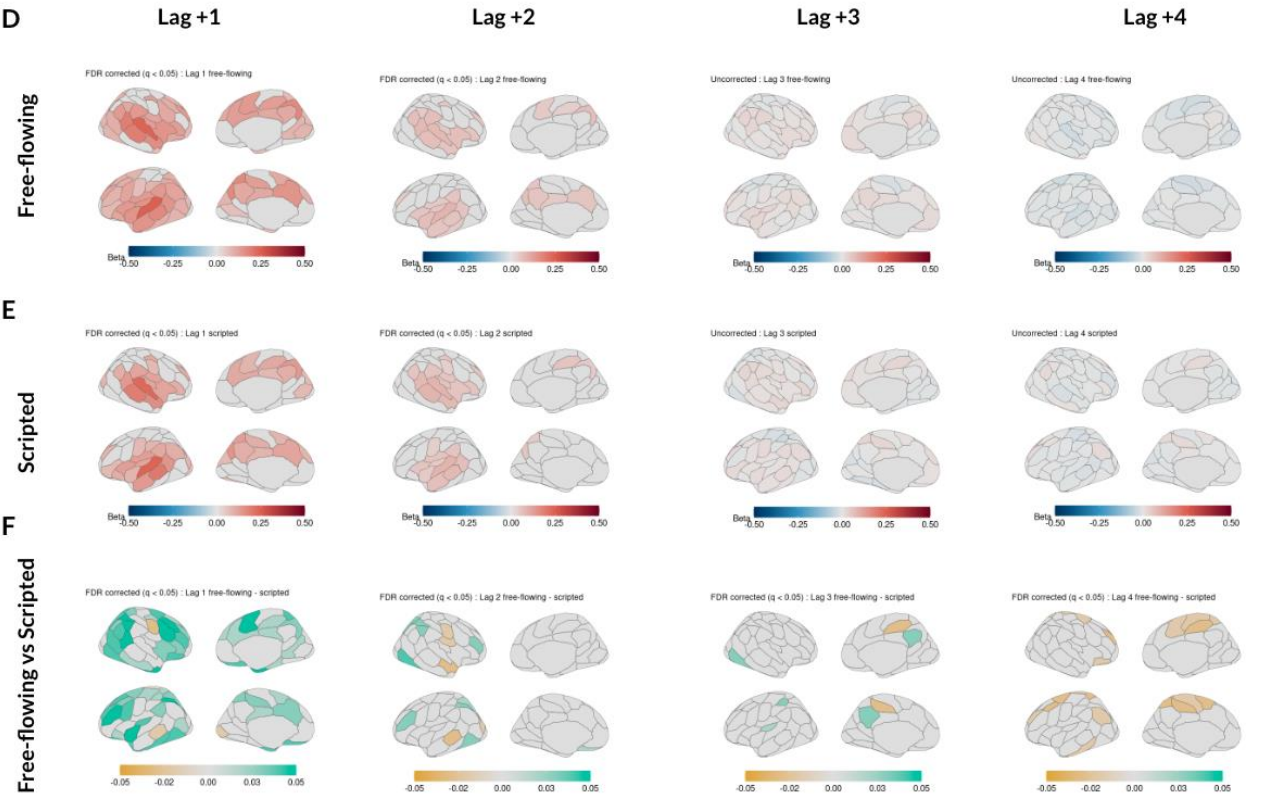


Fig S2. Temporal lag-ISC during the free-flowing, scripted, and free-flowing vs scripted contrast. Shown on the cortical surface is the neural alignment when the (A-C) speaker's activity precedes that of the listener's at time lags starting at -1.5 to -6 seconds (lag -1 to lag -4). (D-F) Neural alignment when the listener's brain activity precedes that of the speaker by 1.4 to 6 seconds (lag +1 to lag +4). In all lag plots depicting the free-flowing vs scripted contrast results, greater alignment during the free-flowing conversations are shown in green, whereas greater alignment during the scripted conversations are depicted in yellow. Results shown on the brain surface that passed multiple corrections are labeled FDR corrected at $q < 0.05$, otherwise they are labeled as uncorrected and the unthresholded results are shown.

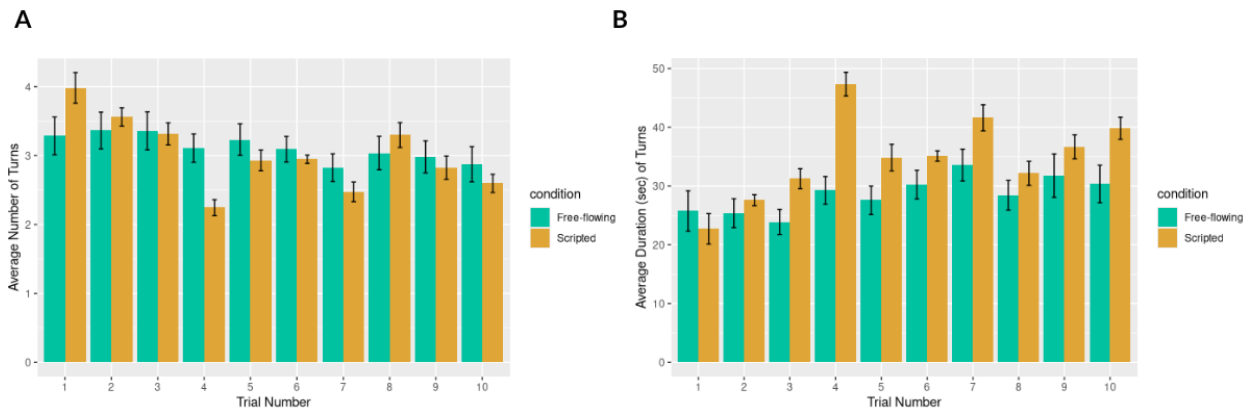


Fig S3. Number of turns and duration of turns during free-flowing and scripted conversation trials. (A) Average number of turns and (B) the average duration of turns in seconds across dyads during both conversation conditions. Only turns that were 14TRs (21 seconds) long were included given that shorter turns were removed because they did not have sufficient enough data to perform the lagged-ISC analysis. Error bars in both plots represent the 95% confidence interval. The overall average of number of turns (free-flowing: $M_{turns} = 3.11$, $SD_{turns} = 0.90$; scripted: $M_{turns} = 3.02$, $SD_{turns} = 0.76$) and duration (free-flowing: $M_{duration} = 28.66$, $SD_{duration} = 10.68$; scripted: $M_{duration} = 34.91$, $SD_{duration} = 9.85$).

Table S1.

Participant demographic

Participant Characteristics		Full sample (n=114)	Friends (n=60)	Strangers (n=54)
Age, mean (SD)		20.6 (3.0)	20.4 (2.3)	20.8 (3.6)
Gender, n (%)	Female	73 (64.0)	36 (60.0)	37 (68.5)
	Male	41 (36.0)	24 (40.0)	17 (31.5)
Length known (months), mean (SD)		10.4 (14.8)	19.7 (15.3)	0.0 (0.0)
Race, n (%)	Asian	40 (35.1)	23 (38.3)	17 (31.5)
	Black and/or African American	10 (8.8)	3 (5.0)	7 (13.0)
	More than one race	28 (24.6)	15 (25.0)	13 (24.1)
	Other	2 (1.8)	1 (1.7)	1 (1.9)
	White	34 (29.8)	18 (30.0)	16 (29.6)
Ethnicity, n (%)	Latino/a/x	22 (19.5)	13 (21.7)	9 (17.0)

Table S2.

List of prompts used in the conversation task.

Set	Item	Question
1	1	Given the choice of anyone in the world, whom would you want as a dinner guest?
1	2	Would you like to be famous? In what way?
1	3	Before making a telephone call, do you ever rehearse what you are going to say? Why?
1	4	What would constitute a "perfect" day for you?
1	5	When did you last sing to yourself? To someone else?
1	6	If you were able to live to the age of 90 and retain either the mind or body of a 30-year-old for the last 60 years of your life, which would you want?
1	7	For what in your life do you feel most grateful?
1	8	If you could change anything about the way you were raised, what would it be?
2	9	If a crystal ball could tell you the truth about yourself, your life, the future, or anything else, what would you want to know?
2	10	Is there something that you've dreamed of doing for a long time? Why haven't you done it?
2	11	What is the greatest accomplishment of your life?
2	12	What do you value most in a friendship?

2	13	What is your most treasured memory?
2	14	How close and warm is your family? Do you feel your childhood was happier than most other people's?
3	15	Complete this sentence: I wish I had someone with whom I could share…
3	16	Please share what would be important for your study partner to know as your close friend.
3	17	Share with your partner an embarrassing moment in your life.
3	18	What, if anything, is too serious to be joked about?
3	19	Your house, containing everything you own, catches fire. After saving your loved ones and pets, you have time to safely make a final dash to save any one item. What would it be? Why?
3	20	Share a personal problem and ask your partner's advice on how he or she might handle it. Also, ask your partner to reflect back to you how you seem to be feeling about the problem you have chosen.

Sets = The three different levels of intimacy of prompts.

Table S3

Time locked ISC regression results.

Parcel	Contrast	Estimate	SE	p-value	p-value (FDR corrected)	17Network Parcel Label
1	Free-flowing - Scripted	-0.012	0.008	0.124	0.182	17Networks_LH_VisCent_ExStr_1
2	Free-flowing - Scripted	0.005	0.008	0.529	0.601	17Networks_LH_VisCent_ExStr_2
3	Free-flowing - Scripted	-0.017	0.008	0.035	0.055	17Networks_LH_VisCent_Striate_1
4	Free-flowing - Scripted	-0.009	0.009	0.318	0.398	17Networks_LH_VisCent_ExStr_3
5	Free-flowing - Scripted	-0.009	0.008	0.275	0.357	17Networks_LH_VisPeri_ExStrInf_1
6	Free-flowing - Scripted	0.009	0.008	0.278	0.357	17Networks_LH_VisPeri_StriCal_1
7	Free-flowing - Scripted	0.011	0.008	0.164	0.228	17Networks_LH_VisPeri_ExStrSup_1
8	Free-flowing - Scripted	0.028	0.007	0	0	17Networks_LH_SomMotA_1
9	Free-flowing - Scripted	0.009	0.008	0.237	0.317	17Networks_LH_SomMotA_2
10	Free-flowing - Scripted	0.012	0.008	0.131	0.19	17Networks_LH_SomMotB_Aud_1
11	Free-flowing - Scripted	0.009	0.008	0.256	0.336	17Networks_LH_SomMotB_S2_1
12	Free-flowing - Scripted	-0.002	0.008	0.849	0.875	17Networks_LH_SomMotB_S2_2
13	Free-flowing - Scripted	-0.008	0.008	0.343	0.423	17Networks_LH_SomMotB_Cent_1
14	Free-flowing - Scripted	0.023	0.008	0.003	0.006	17Networks_LH_DorsAttnA_TempOcc_1

15	Free-flowing - Scripted	0.029	0.008	0	0.001	17Networks_LH_DorsAttnA_ParOcc_1
16	Free-flowing - Scripted	0.041	0.008	0	0	17Networks_LH_DorsAttnA_SPL_1
17	Free-flowing - Scripted	-0.038	0.008	0	0	17Networks_LH_DorsAttnB_PostC_1
18	Free-flowing - Scripted	0.006	0.008	0.421	0.495	17Networks_LH_DorsAttnB_PostC_2
19	Free-flowing - Scripted	0.005	0.008	0.581	0.639	17Networks_LH_DorsAttnB_PostC_3
20	Free-flowing - Scripted	0.02	0.008	0.016	0.028	17Networks_LH_DorsAttnB_FEF_1
21	Free-flowing - Scripted	0.044	0.008	0	0	17Networks_LH_SalVentAttnA_ParOper_1
22	Free-flowing - Scripted	0.066	0.008	0	0	17Networks_LH_SalVentAttnA_Ins_1
23	Free-flowing - Scripted	0.027	0.008	0.001	0.002	17Networks_LH_SalVentAttnA_Ins_2
24	Free-flowing - Scripted	0.043	0.008	0	0	17Networks_LH_SalVentAttnA_ParMed_1
25	Free-flowing - Scripted	0.024	0.008	0.003	0.005	17Networks_LH_SalVentAttnA_FrMed_1
26	Free-flowing - Scripted	0.046	0.009	0	0	17Networks_LH_SalVentAttnB_PFCl_1
27	Free-flowing - Scripted	0.028	0.009	0.001	0.002	17Networks_LH_SalVentAttnB_PFCmp_1
28	Free-flowing - Scripted	0.031	0.008	0	0	17Networks_LH_LimbicB_OFC_1
29	Free-flowing - Scripted	0.032	0.008	0	0	17Networks_LH_LimbicA_TempPole_1
30	Free-flowing - Scripted	0.052	0.008	0	0	17Networks_LH_LimbicA_TempPole_2
31	Free-flowing - Scripted	0.053	0.008	0	0	17Networks_LH_ContA_IPS_1

32	Free-flowing - Scripted	0.036	0.008	0	0	17Networks_LH_ContA_PFCl_1
33	Free-flowing - Scripted	0.025	0.008	0.002	0.004	17Networks_LH_ContA_PFCl_2
34	Free-flowing - Scripted	0.017	0.008	0.034	0.055	17Networks_LH_ContB_PFClv_1
35	Free-flowing - Scripted	0.019	0.008	0.023	0.04	17Networks_LH_ContC_pCun_1
36	Free-flowing - Scripted	0.033	0.008	0	0	17Networks_LH_ContC_pCun_2
37	Free-flowing - Scripted	0.018	0.008	0.031	0.05	17Networks_LH_ContC_Cingp_1
38	Free-flowing - Scripted	0.047	0.008	0	0	17Networks_LH_DefaultA_PFCd_1
39	Free-flowing - Scripted	0.012	0.008	0.145	0.207	17Networks_LH_DefaultA_pCunPCC_1
40	Free-flowing - Scripted	0.007	0.008	0.365	0.44	17Networks_LH_DefaultA_PFCm_1
41	Free-flowing - Scripted	0.009	0.008	0.289	0.366	17Networks_LH_DefaultB_Temp_1
42	Free-flowing - Scripted	-0.032	0.008	0	0	17Networks_LH_DefaultB_Temp_2
43	Free-flowing - Scripted	0.015	0.008	0.056	0.088	17Networks_LH_DefaultB_IPL_1
44	Free-flowing - Scripted	0.025	0.008	0.002	0.004	17Networks_LH_DefaultB_PFCd_1
45	Free-flowing - Scripted	0.026	0.008	0.002	0.004	17Networks_LH_DefaultB_PFCl_1
46	Free-flowing - Scripted	0.026	0.008	0.001	0.002	17Networks_LH_DefaultB_PFCv_1
47	Free-flowing - Scripted	0.006	0.008	0.485	0.564	17Networks_LH_DefaultB_PFCv_2
48	Free-flowing - Scripted	-0.015	0.008	0.066	0.102	17Networks_LH_DefaultC_Rsp_1

49	Free-flowing - Scripted	-0.018	0.008	0.018	0.031	17Networks_LH_DefaultC_PHC_1
50	Free-flowing - Scripted	0.027	0.008	0.001	0.002	17Networks_LH_TempPar_1
51	Free-flowing - Scripted	0.001	0.008	0.951	0.96	17Networks_RH_VisCent_ExStr_1
52	Free-flowing - Scripted	-0.001	0.008	0.887	0.905	17Networks_RH_VisCent_ExStr_2
53	Free-flowing - Scripted	-0.006	0.008	0.506	0.581	17Networks_RH_VisCent_ExStr_3
54	Free-flowing - Scripted	0.007	0.008	0.364	0.44	17Networks_RH_VisPeri_StriCal_1
55	Free-flowing - Scripted	0.021	0.008	0.006	0.01	17Networks_RH_VisPeri_ExStrInf_1
56	Free-flowing - Scripted	-0.003	0.008	0.732	0.778	17Networks_RH_VisPeri_ExStrSup_1
57	Free-flowing - Scripted	0.034	0.008	0	0	17Networks_RH_SomMotA_1
58	Free-flowing - Scripted	-0.032	0.007	0	0	17Networks_RH_SomMotA_2
59	Free-flowing - Scripted	0	0.008	0.985	0.985	17Networks_RH_SomMotA_3
60	Free-flowing - Scripted	0.003	0.008	0.727	0.778	17Networks_RH_SomMotA_4
61	Free-flowing - Scripted	-0.01	0.008	0.237	0.317	17Networks_RH_SomMotB_Aud_1
62	Free-flowing - Scripted	-0.02	0.008	0.017	0.029	17Networks_RH_SomMotB_S2_1
63	Free-flowing - Scripted	0.003	0.008	0.76	0.8	17Networks_RH_SomMotB_S2_2
64	Free-flowing - Scripted	-0.026	0.008	0.001	0.003	17Networks_RH_SomMotB_Cent_1
65	Free-flowing - Scripted	0.03	0.008	0	0.001	17Networks_RH_DorsAttnA_TempOcc_1

66	Free-flowing - Scripted	0.044	0.008	0	0	17Networks_RH_DorsAttnA_ParOcc_1
67	Free-flowing - Scripted	0.034	0.008	0	0	17Networks_RH_DorsAttnA_SPL_1
68	Free-flowing - Scripted	-0.011	0.008	0.153	0.215	17Networks_RH_DorsAttnB_PostC_1
69	Free-flowing - Scripted	0.024	0.008	0.003	0.005	17Networks_RH_DorsAttnB_PostC_2
70	Free-flowing - Scripted	0.036	0.008	0	0	17Networks_RH_DorsAttnB_FEF_1
71	Free-flowing - Scripted	0.002	0.008	0.79	0.823	17Networks_RH_SalVentAttnA_ParOper_1
72	Free-flowing - Scripted	0.028	0.008	0.001	0.002	17Networks_RH_SalVentAttnA_Ins_1
73	Free-flowing - Scripted	0.043	0.008	0	0	17Networks_RH_SalVentAttnA_ParMed_1
74	Free-flowing - Scripted	0.062	0.009	0	0	17Networks_RH_SalVentAttnA_FrMed_1
75	Free-flowing - Scripted	0.061	0.008	0	0	17Networks_RH_SalVentAttnB_IPL_1
76	Free-flowing - Scripted	0.058	0.008	0	0	17Networks_RH_SalVentAttnB_PFCl_1
77	Free-flowing - Scripted	0.028	0.008	0.001	0.002	17Networks_RH_SalVentAttnB_PFCmp_1
78	Free-flowing - Scripted	0.037	0.008	0	0	17Networks_RH_LimbicB_OFC_1
79	Free-flowing - Scripted	0.045	0.008	0	0	17Networks_RH_LimbicA_TempPole_1
80	Free-flowing - Scripted	0.055	0.008	0	0	17Networks_RH_ContA_IPS_1
81	Free-flowing - Scripted	0.038	0.008	0	0	17Networks_RH_ContA_PFCl_1
82	Free-flowing - Scripted	0.057	0.008	0	0	17Networks_RH_ContA_PFCl_2

83	Free-flowing - Scripted	0.024	0.008	0.002	0.004	17Networks_RH_ContB_Temp_1
84	Free-flowing - Scripted	0.011	0.008	0.19	0.261	17Networks_RH_ContB_IPL_1
85	Free-flowing - Scripted	0.053	0.008	0	0	17Networks_RH_ContB_PFCld_1
86	Free-flowing - Scripted	0.029	0.008	0.001	0.002	17Networks_RH_ContB_PFClv_1
87	Free-flowing - Scripted	0.032	0.009	0	0.001	17Networks_RH_ContC_Cingp_1
88	Free-flowing - Scripted	0.019	0.009	0.026	0.043	17Networks_RH_ContC_pCun_1
89	Free-flowing - Scripted	0.045	0.008	0	0	17Networks_RH_DefaultA_IPL_1
90	Free-flowing - Scripted	0.032	0.008	0	0	17Networks_RH_DefaultA_PFCd_1
91	Free-flowing - Scripted	0.013	0.008	0.107	0.16	17Networks_RH_DefaultA_pCunPCC_1
92	Free-flowing - Scripted	0.005	0.008	0.56	0.629	17Networks_RH_DefaultA_PFCm_1
93	Free-flowing - Scripted	0.045	0.008	0	0	17Networks_RH_DefaultB_PFCd_1
94	Free-flowing - Scripted	0.046	0.008	0	0	17Networks_RH_DefaultB_PFCv_1
95	Free-flowing - Scripted	0.034	0.008	0	0	17Networks_RH_DefaultB_PFCv_2
96	Free-flowing - Scripted	-0.007	0.008	0.372	0.443	17Networks_RH_DefaultC_Rsp_1
97	Free-flowing - Scripted	0.004	0.008	0.57	0.634	17Networks_RH_DefaultC_PHC_1
98	Free-flowing - Scripted	0.014	0.008	0.077	0.116	17Networks_RH_TempPar_1
99	Free-flowing - Scripted	0.004	0.008	0.599	0.651	17Networks_RH_TempPar_2

100	Free-flowing - Scripted	0.074	0.008	0	0	17Networks_RH_TempPar_3

Parcel	Contrast	Estimate	SE	p-value	p-value (FDR corrected)	17Network Parcel Label
1	Scripted	0.223	0.077	0.004	0.009	17Networks_LH_VisCent_ExStr_1
2	Scripted	0.195	0.067	0.004	0.008	17Networks_LH_VisCent_ExStr_2
3	Scripted	0.185	0.076	0.015	0.023	17Networks_LH_VisCent_Striate_1
4	Scripted	0.238	0.071	0.001	0.003	17Networks_LH_VisCent_ExStr_3
5	Scripted	0.18	0.068	0.008	0.014	17Networks_LH_VisPeri_ExStrInf_1
6	Scripted	0.164	0.069	0.018	0.026	17Networks_LH_VisPeri_StriCal_1
7	Scripted	0.229	0.071	0.001	0.004	17Networks_LH_VisPeri_ExStrSup_1
8	Scripted	0.063	0.071	0.372	0.372	17Networks_LH_SomMotA_1
9	Scripted	0.148	0.081	0.069	0.083	17Networks_LH_SomMotA_2
10	Scripted	0.684	0.069	0	0	17Networks_LH_SomMotB_Aud_1
11	Scripted	0.323	0.068	0	0	17Networks_LH_SomMotB_S2_1
12	Scripted	0.331	0.073	0	0	17Networks_LH_SomMotB_S2_2
13	Scripted	0.166	0.065	0.01	0.017	17Networks_LH_SomMotB_Cent_1
14	Scripted	0.189	0.074	0.01	0.017	17Networks_LH_DorsAttnA_TempOcc_1

15	Scripted	0.23	0.08	0.004	0.009	17Networks_LH_DorsAttnA_ParOcc_1
16	Scripted	0.187	0.08	0.019	0.028	17Networks_LH_DorsAttnA_SPL_1
17	Scripted	0.208	0.056	0	0.001	17Networks_LH_DorsAttnB_PostC_1
18	Scripted	0.097	0.07	0.165	0.174	17Networks_LH_DorsAttnB_PostC_2
19	Scripted	0.168	0.064	0.009	0.015	17Networks_LH_DorsAttnB_PostC_3
20	Scripted	0.17	0.077	0.026	0.036	17Networks_LH_DorsAttnB_FEF_1
21	Scripted	0.235	0.066	0	0.001	17Networks_LH_SalVentAttnA_ParOper_1
22	Scripted	0.278	0.068	0	0	17Networks_LH_SalVentAttnA_Ins_1
23	Scripted	0.278	0.065	0	0	17Networks_LH_SalVentAttnA_Ins_2
24	Scripted	0.218	0.078	0.005	0.01	17Networks_LH_SalVentAttnA_ParMed_1
25	Scripted	0.24	0.071	0.001	0.002	17Networks_LH_SalVentAttnA_FrMed_1
26	Scripted	0.202	0.072	0.005	0.01	17Networks_LH_SalVentAttnB_PFCl_1
27	Scripted	0.267	0.074	0	0.001	17Networks_LH_SalVentAttnB_PFCmp_1
28	Scripted	0.085	0.06	0.156	0.166	17Networks_LH_LimbicB_OFC_1
29	Scripted	0.196	0.066	0.003	0.007	17Networks_LH_LimbicA_TempPole_1
30	Scripted	0.224	0.066	0.001	0.002	17Networks_LH_LimbicA_TempPole_2
31	Scripted	0.203	0.071	0.004	0.009	17Networks_LH_ContA_IPS_1

32	Scripted	0.141	0.063	0.024	0.034	17Networks_LH_ContA_PFCl_1
33	Scripted	0.22	0.065	0.001	0.002	17Networks_LH_ContA_PFCl_2
34	Scripted	0.087	0.067	0.198	0.204	17Networks_LH_ContB_PFClv_1
35	Scripted	0.393	0.074	0	0	17Networks_LH_ContC_pCun_1
36	Scripted	0.344	0.08	0	0	17Networks_LH_ContC_pCun_2
37	Scripted	0.219	0.07	0.002	0.005	17Networks_LH_ContC_Cingp_1
38	Scripted	0.178	0.071	0.013	0.02	17Networks_LH_DefaultA_PFCd_1
39	Scripted	0.256	0.081	0.002	0.004	17Networks_LH_DefaultA_pCunPCC_1
40	Scripted	0.115	0.068	0.091	0.104	17Networks_LH_DefaultA_PFCm_1
41	Scripted	0.388	0.077	0	0	17Networks_LH_DefaultB_Temp_1
42	Scripted	0.481	0.082	0	0	17Networks_LH_DefaultB_Temp_2
43	Scripted	0.264	0.082	0.001	0.004	17Networks_LH_DefaultB_IPL_1
44	Scripted	0.205	0.073	0.005	0.01	17Networks_LH_DefaultB_PFCd_1
45	Scripted	0.206	0.074	0.006	0.01	17Networks_LH_DefaultB_PFCl_1
46	Scripted	0.262	0.06	0	0	17Networks_LH_DefaultB_PFCv_1
47	Scripted	0.195	0.058	0.001	0.003	17Networks_LH_DefaultB_PFCv_2
48	Scripted	0.167	0.073	0.022	0.032	17Networks_LH_DefaultC_Rsp_1

49	Scripted	0.137	0.07	0.052	0.064	17Networks_LH_DefaultC_PHC_1
50	Scripted	0.382	0.074	0	0	17Networks_LH_TempPar_1
51	Scripted	0.201	0.076	0.008	0.014	17Networks_RH_VisCent_ExStr_1
52	Scripted	0.198	0.07	0.005	0.009	17Networks_RH_VisCent_ExStr_2
53	Scripted	0.222	0.074	0.003	0.007	17Networks_RH_VisCent_ExStr_3
54	Scripted	0.202	0.071	0.005	0.009	17Networks_RH_VisPeri_StriCal_1
55	Scripted	0.19	0.069	0.006	0.011	17Networks_RH_VisPeri_ExStrInf_1
56	Scripted	0.242	0.076	0.001	0.004	17Networks_RH_VisPeri_ExStrSup_1
57	Scripted	0.255	0.064	0	0	17Networks_RH_SomMotA_1
58	Scripted	0.105	0.065	0.104	0.116	17Networks_RH_SomMotA_2
59	Scripted	0.116	0.069	0.094	0.107	17Networks_RH_SomMotA_3
60	Scripted	0.147	0.072	0.041	0.054	17Networks_RH_SomMotA_4
61	Scripted	0.654	0.067	0	0	17Networks_RH_SomMotB_Aud_1
62	Scripted	0.287	0.068	0	0	17Networks_RH_SomMotB_S2_1
63	Scripted	0.261	0.062	0	0	17Networks_RH_SomMotB_S2_2
64	Scripted	0.141	0.056	0.012	0.019	17Networks_RH_SomMotB_Cent_1
65	Scripted	0.181	0.074	0.014	0.022	17Networks_RH_DorsAttnA_TempOcc_1

66	Scripted	0.193	0.086	0.025	0.034	17Networks_RH_DorsAttnA_ParOcc_1
67	Scripted	0.184	0.081	0.024	0.034	17Networks_RH_DorsAttnA_SPL_1
68	Scripted	0.155	0.071	0.029	0.039	17Networks_RH_DorsAttnB_PostC_1
69	Scripted	0.185	0.071	0.009	0.015	17Networks_RH_DorsAttnB_PostC_2
70	Scripted	0.129	0.07	0.066	0.081	17Networks_RH_DorsAttnB_FEF_1
71	Scripted	0.251	0.06	0	0	17Networks_RH_SalVentAttnA_ParOper_1
72	Scripted	0.262	0.065	0	0	17Networks_RH_SalVentAttnA_Ins_1
73	Scripted	0.213	0.07	0.002	0.006	17Networks_RH_SalVentAttnA_ParMed_1
74	Scripted	0.236	0.068	0	0.002	17Networks_RH_SalVentAttnA_FrMed_1
75	Scripted	0.12	0.07	0.086	0.103	17Networks_RH_SalVentAttnB_IPL_1
76	Scripted	0.218	0.068	0.001	0.004	17Networks_RH_SalVentAttnB_PFCl_1
77	Scripted	0.224	0.075	0.003	0.007	17Networks_RH_SalVentAttnB_PFCmp_1
78	Scripted	0.067	0.069	0.335	0.339	17Networks_RH_LimbicB_OFC_1
79	Scripted	0.128	0.063	0.042	0.054	17Networks_RH_LimbicA_TempPole_1
80	Scripted	0.108	0.075	0.154	0.166	17Networks_RH_ContA_IPS_1
81	Scripted	0.086	0.059	0.144	0.156	17Networks_RH_ContA_PFCl_1
82	Scripted	0.14	0.072	0.051	0.064	17Networks_RH_ContA_PFCl_2

83	Scripted	0.115	0.086	0.182	0.189	17Networks_RH_ContB_Temp_1
84	Scripted	0.154	0.079	0.051	0.064	17Networks_RH_ContB_IPL_1
85	Scripted	0.136	0.079	0.088	0.103	17Networks_RH_ContB_PFCld_1
86	Scripted	0.11	0.07	0.119	0.131	17Networks_RH_ContB_PFClv_1
87	Scripted	0.209	0.069	0.003	0.006	17Networks_RH_ContC_Cingp_1
88	Scripted	0.372	0.073	0	0	17Networks_RH_ContC_pCun_1
89	Scripted	0.24	0.088	0.006	0.011	17Networks_RH_DefaultA_IPL_1
90	Scripted	0.165	0.078	0.033	0.044	17Networks_RH_DefaultA_PFCd_1
91	Scripted	0.259	0.082	0.002	0.004	17Networks_RH_DefaultA_pCunPCC_1
92	Scripted	0.114	0.07	0.101	0.114	17Networks_RH_DefaultA_PFCm_1
93	Scripted	0.201	0.084	0.017	0.025	17Networks_RH_DefaultB_PFCd_1
94	Scripted	0.195	0.065	0.003	0.006	17Networks_RH_DefaultB_PFCv_1
95	Scripted	0.211	0.059	0	0.001	17Networks_RH_DefaultB_PFCv_2
96	Scripted	0.122	0.072	0.09	0.104	17Networks_RH_DefaultC_Rsp_1
97	Scripted	0.09	0.071	0.206	0.211	17Networks_RH_DefaultC_PHC_1
98	Scripted	0.414	0.074	0	0	17Networks_RH_TempPar_1
99	Scripted	0.484	0.082	0	0	17Networks_RH_TempPar_2

100	Scripted	0.363	0.073	0	0	17Networks_RH_TempPar_3

Parcel	Contrast	Estimate	SE	p-value	p-value (FDR corrected)	17Network Parcel Label
1	Free-flowing	0.211	0.077	0.007	0.01	17Networks_LH_VisCent_ExStr_1
2	Free-flowing	0.2	0.067	0.003	0.006	17Networks_LH_VisCent_ExStr_2
3	Free-flowing	0.168	0.076	0.027	0.035	17Networks_LH_VisCent_Striate_1
4	Free-flowing	0.229	0.071	0.001	0.003	17Networks_LH_VisCent_ExStr_3
5	Free-flowing	0.171	0.068	0.012	0.017	17Networks_LH_VisPeri_ExStrInf_1
6	Free-flowing	0.173	0.069	0.013	0.017	17Networks_LH_VisPeri_StriCal_1
7	Free-flowing	0.24	0.071	0.001	0.002	17Networks_LH_VisPeri_ExStrSup_1
8	Free-flowing	0.091	0.071	0.196	0.198	17Networks_LH_SomMotA_1
9	Free-flowing	0.157	0.081	0.053	0.06	17Networks_LH_SomMotA_2
10	Free-flowing	0.696	0.069	0	0	17Networks_LH_SomMotB_Aud_1
11	Free-flowing	0.332	0.068	0	0	17Networks_LH_SomMotB_S2_1
12	Free-flowing	0.33	0.073	0	0	17Networks_LH_SomMotB_S2_2
13	Free-flowing	0.158	0.065	0.014	0.019	17Networks_LH_SomMotB_Cent_1
14	Free-flowing	0.212	0.074	0.004	0.007	17Networks_LH_DorsAttnA_TempOcc_1

15	Free-flowing	0.259	0.08	0.001	0.003	17Networks_LH_DorsAttnA_ParOcc_1
16	Free-flowing	0.228	0.08	0.004	0.007	17Networks_LH_DorsAttnA_SPL_1
17	Free-flowing	0.17	0.056	0.003	0.005	17Networks_LH_DorsAttnB_PostC_1
18	Free-flowing	0.103	0.07	0.139	0.143	17Networks_LH_DorsAttnB_PostC_2
19	Free-flowing	0.173	0.064	0.007	0.011	17Networks_LH_DorsAttnB_PostC_3
20	Free-flowing	0.19	0.077	0.013	0.018	17Networks_LH_DorsAttnB_FEF_1
21	Free-flowing	0.279	0.065	0	0	17Networks_LH_SalVentAttnA_ParOper_1
22	Free-flowing	0.344	0.068	0	0	17Networks_LH_SalVentAttnA_Ins_1
23	Free-flowing	0.305	0.065	0	0	17Networks_LH_SalVentAttnA_Ins_2
24	Free-flowing	0.261	0.078	0.001	0.002	17Networks_LH_SalVentAttnA_ParMed_1
25	Free-flowing	0.264	0.071	0	0.001	17Networks_LH_SalVentAttnA_FrMed_1
26	Free-flowing	0.248	0.072	0.001	0.002	17Networks_LH_SalVentAttnB_PFCl_1
27	Free-flowing	0.296	0.074	0	0	17Networks_LH_SalVentAttnB_PFCmp_1
28	Free-flowing	0.116	0.06	0.053	0.06	17Networks_LH_LimbicB_OFC_1
29	Free-flowing	0.228	0.066	0.001	0.001	17Networks_LH_LimbicA_TempPole_1
30	Free-flowing	0.276	0.066	0	0	17Networks_LH_LimbicA_TempPole_2
31	Free-flowing	0.256	0.071	0	0.001	17Networks_LH_ContA_IPS_1

32	Free-flowing	0.177	0.063	0.005	0.008	17Networks_LH_ContA_PFCl_1
33	Free-flowing	0.244	0.065	0	0.001	17Networks_LH_ContA_PFCl_2
34	Free-flowing	0.104	0.067	0.122	0.128	17Networks_LH_ContB_PFClv_1
35	Free-flowing	0.411	0.074	0	0	17Networks_LH_ContC_pCun_1
36	Free-flowing	0.376	0.08	0	0	17Networks_LH_ContC_pCun_2
37	Free-flowing	0.238	0.07	0.001	0.002	17Networks_LH_ContC_Cingp_1
38	Free-flowing	0.225	0.071	0.002	0.003	17Networks_LH_DefaultA_PFCd_1
39	Free-flowing	0.268	0.081	0.001	0.002	17Networks_LH_DefaultA_pCunPCC_1
40	Free-flowing	0.122	0.068	0.072	0.081	17Networks_LH_DefaultA_PFCm_1
41	Free-flowing	0.397	0.077	0	0	17Networks_LH_DefaultB_Temp_1
42	Free-flowing	0.448	0.082	0	0	17Networks_LH_DefaultB_Temp_2
43	Free-flowing	0.279	0.082	0.001	0.002	17Networks_LH_DefaultB_IPL_1
44	Free-flowing	0.231	0.073	0.002	0.003	17Networks_LH_DefaultB_PFCd_1
45	Free-flowing	0.232	0.074	0.002	0.004	17Networks_LH_DefaultB_PFCl_1
46	Free-flowing	0.288	0.06	0	0	17Networks_LH_DefaultB_PFCv_1
47	Free-flowing	0.201	0.058	0.001	0.001	17Networks_LH_DefaultB_PFCv_2
48	Free-flowing	0.153	0.073	0.037	0.045	17Networks_LH_DefaultC_Rsp_1

49	Free-flowing	0.119	0.07	0.091	0.1	17Networks_LH_DefaultC_PHC_1
50	Free-flowing	0.409	0.074	0	0	17Networks_LH_TempPar_1
51	Free-flowing	0.202	0.076	0.008	0.011	17Networks_RH_VisCent_ExStr_1
52	Free-flowing	0.196	0.07	0.005	0.008	17Networks_RH_VisCent_ExStr_2
53	Free-flowing	0.216	0.074	0.004	0.006	17Networks_RH_VisCent_ExStr_3
54	Free-flowing	0.209	0.071	0.003	0.006	17Networks_RH_VisPeri_StriCal_1
55	Free-flowing	0.211	0.069	0.002	0.005	17Networks_RH_VisPeri_ExStrInf_1
56	Free-flowing	0.239	0.076	0.002	0.003	17Networks_RH_VisPeri_ExStrSup_1
57	Free-flowing	0.29	0.064	0	0	17Networks_RH_SomMotA_1
58	Free-flowing	0.073	0.065	0.26	0.26	17Networks_RH_SomMotA_2
59	Free-flowing	0.116	0.069	0.094	0.103	17Networks_RH_SomMotA_3
60	Free-flowing	0.15	0.072	0.038	0.045	17Networks_RH_SomMotA_4
61	Free-flowing	0.644	0.067	0	0	17Networks_RH_SomMotB_Aud_1
62	Free-flowing	0.267	0.068	0	0	17Networks_RH_SomMotB_S2_1
63	Free-flowing	0.264	0.062	0	0	17Networks_RH_SomMotB_S2_2
64	Free-flowing	0.115	0.056	0.041	0.048	17Networks_RH_SomMotB_Cent_1
65	Free-flowing	0.211	0.074	0.004	0.007	17Networks_RH_DorsAttnA_TempOcc_1

66	Free-flowing	0.237	0.086	0.006	0.009	17Networks_RH_DorsAttnA_ParOcc_1
67	Free-flowing	0.218	0.081	0.008	0.011	17Networks_RH_DorsAttnA_SPL_1
68	Free-flowing	0.144	0.071	0.043	0.051	17Networks_RH_DorsAttnB_PostC_1
69	Free-flowing	0.209	0.071	0.003	0.006	17Networks_RH_DorsAttnB_PostC_2
70	Free-flowing	0.165	0.07	0.019	0.025	17Networks_RH_DorsAttnB_FEF_1
71	Free-flowing	0.253	0.06	0	0	17Networks_RH_SalVentAttnA_ParOper_1
72	Free-flowing	0.29	0.065	0	0	17Networks_RH_SalVentAttnA_Ins_1
73	Free-flowing	0.256	0.07	0	0.001	17Networks_RH_SalVentAttnA_ParMed_1
74	Free-flowing	0.298	0.068	0	0	17Networks_RH_SalVentAttnA_FrMed_1
75	Free-flowing	0.181	0.07	0.01	0.014	17Networks_RH_SalVentAttnB_IPL_1
76	Free-flowing	0.277	0.068	0	0	17Networks_RH_SalVentAttnB_PFCl_1
77	Free-flowing	0.252	0.075	0.001	0.002	17Networks_RH_SalVentAttnB_PFCmp_1
78	Free-flowing	0.104	0.069	0.135	0.141	17Networks_RH_LimbicB_OFC_1
79	Free-flowing	0.173	0.063	0.006	0.009	17Networks_RH_LimbicA_TempPole_1
80	Free-flowing	0.163	0.075	0.031	0.04	17Networks_RH_ContA_IPS_1
81	Free-flowing	0.124	0.059	0.035	0.044	17Networks_RH_ContA_PFCl_1
82	Free-flowing	0.197	0.072	0.006	0.009	17Networks_RH_ContA_PFCl_2

83	Free-flowing	0.139	0.086	0.106	0.113	17Networks_RH_ContB_Temp_1
84	Free-flowing	0.165	0.079	0.037	0.045	17Networks_RH_ContB_IPL_1
85	Free-flowing	0.188	0.079	0.018	0.023	17Networks_RH_ContB_PFCld_1
86	Free-flowing	0.139	0.07	0.049	0.057	17Networks_RH_ContB_PFClv_1
87	Free-flowing	0.241	0.069	0	0.001	17Networks_RH_ContC_Cingp_1
88	Free-flowing	0.391	0.073	0	0	17Networks_RH_ContC_pCun_1
89	Free-flowing	0.285	0.088	0.001	0.003	17Networks_RH_DefaultA_IPL_1
90	Free-flowing	0.197	0.078	0.011	0.015	17Networks_RH_DefaultA_PFCd_1
91	Free-flowing	0.272	0.082	0.001	0.002	17Networks_RH_DefaultA_pCunPCC_1
92	Free-flowing	0.119	0.07	0.088	0.098	17Networks_RH_DefaultA_PFCm_1
93	Free-flowing	0.246	0.084	0.004	0.006	17Networks_RH_DefaultB_PFCd_1
94	Free-flowing	0.241	0.065	0	0.001	17Networks_RH_DefaultB_PFCv_1
95	Free-flowing	0.245	0.059	0	0	17Networks_RH_DefaultB_PFCv_2
96	Free-flowing	0.115	0.072	0.11	0.117	17Networks_RH_DefaultC_Rsp_1
97	Free-flowing	0.095	0.071	0.185	0.189	17Networks_RH_DefaultC_PHC_1
98	Free-flowing	0.427	0.074	0	0	17Networks_RH_TempPar_1
99	Free-flowing	0.488	0.082	0	0	17Networks_RH_TempPar_2

100	Free-flowing	0.436	0.073	0	0	17Networks_RH_TempPar_3

Table S4.

Results of linear mixed effects model using parcel network assignment and condition to predict temporal neural alignment during conversation. Trial and turn duration are included as covariates of no interest.

Effects	**df**	**F**	***p***
Trial	1, 344130.27	0.18	0.68
Duration	1, 344145.89	0.46	0.50
Network	6, 344128.00	287.88	< 0.001
Condition	1, 344132.31	145.73	< 0.001
Network x Condition	6, 344128.00	20.55	< 0.001

Degrees of freedom (df) were calculated using the Satterthwaite approximation.

Table S5.

Post-hoc network by condition interaction.

Free-flowing - Scripted				
Network	Estimate	SE	z	*p-value*
Visual	-0.006	0.004	-1.47	0.14
Somatomotor	-0.002	0.004	-0.49	0.62
Dorsal Attention	**0.02**	**0.004**	**4.07**	**< 0.0001**
Ventral Attention	**0.05**	**0.005**	**10.11**	**< 0.0001**
Limbic	**0.04**	**0.007**	**5.73**	**< 0.0001**
Frontoparietal	**0.03**	**0.004**	**7.30**	**< 0.0001**
Default	**0.02**	**0.003**	**7.23**	**<0.0001**

Table S6.

Pairwise comparison of alignment between networks during free-flowing and scripted conversations.

Pairwise Comparisons

Condition	Contrast	Estimate	SE	p-value
Free-flowing	Visual - Somatomotor	-0.07	0.004	0.000****
Free-flowing	Visual - Dorsal Attention	0.00	0.004	0.271
Free-flowing	Visual - Ventral Attention	-0.11	0.004	0.000****
Free-flowing	Visual - Limbic	0.01	0.006	0.188
Free-flowing	Visual - Frontoparietal Control	-0.04	0.004	0.000****
Free-flowing	Visual - Default	-0.07	0.004	0.000****
Free-flowing	Somatomotor - Dorsal Attention	0.06	0.004	0.000****
Free-flowing	Somatomotor - Ventral Attention	-0.04	0.005	0.000****
Free-flowing	Somatomotor - Limbic	0.07	0.006	0.000****
Free-flowing	Somatomotor - Frontoparietal Control	0.03	0.004	0.000****
Free-flowing	Somatomotor - Default	0.00	0.004	0.395

Free-flowing	Dorsal Attention - Ventral Attention	-0.10	0.004	0.000****
Free-flowing	Dorsal Attention - Limbic	0.01	0.006	0.039*
Free-flowing	Dorsal Attention - Frontoparietal Control	-0.04	0.004	0.000****
Free-flowing	Dorsal Attention - Default	-0.06	0.004	0.000****
Free-flowing	Ventral Attention - Limbic	0.11	0.006	0.000****
Free-flowing	Ventral Attention - Frontoparietal Control	0.07	0.005	0.000****
Free-flowing	Ventral Attention - Default	0.04	0.004	0.000****
Free-flowing	Limbic - Frontoparietal Control	-0.05	0.006	0.000****
Free-flowing	Limbic - Default	-0.08	0.006	0.000****
Free-flowing	Frontoparietal Control - Default	-0.03	0.004	0.000****
Scripted	Visual - Somatomotor	-0.06	0.004	0.000****
Scripted	Visual - Dorsal Attention	0.02	0.004	0.000****
Scripted	Visual - Ventral Attention	-0.05	0.004	0.000****
Scripted	Visual - Limbic	0.06	0.006	0.000****

Scripted	Visual - Frontoparietal Control	0.00	0.004	0.905
Scripted	Visual - Default	-0.04	0.004	0.000****
Scripted	Somatomotor - Dorsal Attention	0.08	0.004	0.000****
Scripted	Somatomotor - Ventral Attention	0.01	0.005	0.025*
Scripted	Somatomotor - Limbic	0.12	0.006	0.000****
Scripted	Somatomotor - Frontoparietal Control	0.06	0.004	0.000****
Scripted	Somatomotor - Default	0.02	0.004	0.000****
Scripted	Dorsal Attention - Ventral Attention	-0.07	0.004	0.000****
Scripted	Dorsal Attention - Limbic	0.04	0.006	0.000****
Scripted	Dorsal Attention - Frontoparietal Control	-0.02	0.004	0.000****
Scripted	Dorsal Attention - Default	-0.06	0.004	0.000****
Scripted	Ventral Attention - Limbic	0.11	0.006	0.000****
Scripted	Ventral Attention - Frontoparietal Control	0.05	0.005	0.000****
Scripted	Ventral Attention - Default	0.01	0.004	0.002**

Scripted	Limbic - Frontoparietal Control	-0.06	0.006	0.000****
Scripted	Limbic - Default	-0.10	0.006	0.000****
Scripted	Frontoparietal Control - Default	-0.04	0.004	0.000****

Note: * p < .05, ** p < .01, *** p < .001., **** p < .0001.

References

Anders, S., Verrel, J., Haynes, J.-D., & Ethofer, T. (2020). Pseudo-hyperscanning shows common neural activity during face-to-face communication of affect to be associated with shared affective feelings but not with mere emotion recognition. *Cortex*, *131*, 210–220. https://doi.org/10.1016/j.cortex.2020.06.015

Aron, A., Melinat, E., Aron, E. N., Vallone, R. D., & Bator, R. J. (1997). The Experimental Generation of Interpersonal Closeness: A Procedure and Some Preliminary Findings. *Personality and Social Psychology Bulletin*, *23*(4), 363–377. https://doi.org/10.1177/0146167297234003

Bazeille, T., DuPre, E., Richard, H., Poline, J.-B., & Thirion, B. (2021). An empirical evaluation of functional alignment using inter-subject decoding. *NeuroImage*, *245*, 118683. https://doi.org/10.1016/j.neuroimage.2021.118683

Bevilacqua, D., Davidesco, I., Wan, L., Chaloner, K., Rowland, J., Ding, M., Poeppel, D., & Dikker, S. (2019). Brain-to-Brain Synchrony and Learning Outcomes Vary by Student-Teacher Dynamics: Evidence from a Real-world Classroom Electroencephalography Study. *Journal of Cognitive Neuroscience*, *31*(3), 401–411.

Burns, S. M., Tsoi, L., Falk, E. B., Speer, S. P. H., Mwilambwe-Tshilobo, L., & Tamir, D. I. (2025). Interdependent minds: Quantifying the dynamics of successful social interaction. *Current Directions in Psychological Science*. https://doi.org/10.1177/09637214251323598

Chen, J., Leong, Y. C., Honey, C. J., Yong, C. H., Norman, K. A., & Hasson, U. (2017a). Shared memories reveal shared structure in neural activity across individuals. *Nature Neuroscience*, *20*(1), 115–125. https://doi.org/10.1038/nn.4450

Chen, J., Leong, Y. C., Honey, C. J., Yong, C. H., Norman, K. A., & Hasson, U. (2017b). Shared memories reveal shared structure in neural activity across individuals. *Nature Neuroscience*, *20*(1), Article 1. https://doi.org/10.1038/nn.4450

Chen, P.-H. (Cameron), Chen, J., Yeshurun, Y., Hasson, U., Haxby, J., & Ramadge, P. J. (2015). A Reduced-Dimension fMRI Shared Response Model. *Advances in Neural Information Processing Systems*, *28*. https://papers.nips.cc/paper_files/paper/2015/hash/b3967a0e938dc2a6340e258630febd5a-Abstract.html

Chen, P.-H., Guntupalli, J. S., Haxby, J. V., & Ramadge, P. J. (2014). Joint SVD-Hyperalignment for multi-subject FMRI data alignment. *2014 IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, 1–6. https://doi.org/10.1109/MLSP.2014.6958912

Dikker, S., Silbert, L. J., Hasson, U., & Zevin, J. D. (2014). On the Same Wavelength: Predictable Language Enhances Speaker–Listener Brain-to-Brain Synchrony in Posterior Superior Temporal Gyrus. *Journal of Neuroscience*, *34*(18), 6267–6272. https://doi.org/10.1523/JNEUROSCI.3796-13.2014

Dikker, S., Wan, L., Davidesco, I., Kaggen, L., Oostrik, M., McClintock, J., Rowland, J., Michalareas, G., Van Bavel, J. J., Ding, M., & Poeppel, D. (2017). Brain-to-Brain Synchrony Tracks Real-World Dynamic Group Interactions in the Classroom. *Current Biology*, *27*(9), 1375–1380. https://doi.org/10.1016/j.cub.2017.04.002

Fedorenko, E., Ivanova, A. A., & Regev, T. I. (2024). The language network as a natural kind within the broader landscape of the human brain. *Nature Reviews Neuroscience*, *25*(5), 289–312. https://doi.org/10.1038/s41583-024-00802-4

Feldman, R., Magori-Cohen, R., Galili, G., Singer, M., & Louzoun, Y. (2011). Mother and infant coordinate heart rhythms through episodes of interaction synchrony. *Infant Behavior & Development*, *34*(4), 569–577.

Garrod, S., & Pickering, M. J. (2004). Why is conversation so easy. *Trends in Cognitive Sciences*, *8*(1), 8–11.

Gorman, K. S., Gegg-Harrison, W., Marsh, C. R., & Tanenhaus, M. K. (2013). What is learned together stays together: Speakers' choice of referring expression reflects

shared experience. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *39*(3), 843–853.

Hamilton, A. F. de C. (2021). Hyperscanning: Beyond the Hype. *Neuron*, *109*(3), 404–407. https://doi.org/10.1016/j.neuron.2020.11.008

Hasson, U., Ghazanfar, A. A., Galantucci, B., Garrod, S., & Keysers, C. (2012). Brain-to-brain coupling: A mechanism for creating and sharing a social world. *Trends in Cognitive Sciences*, *16*(2), 114–121.

Haxby, J. V., Gobbini, M. I., Furey, M. L., Ishai, A., Schouten, J. L., & Pietrini, P. (2001). Distributed and Overlapping Representations of Faces and Objects in Ventral Temporal Cortex. *Science*, *293*(5539), 2425–2430. https://doi.org/10.1126/science.1063736

Jenkinson, M., Bannister, P., Brady, M., & Smith, S. (2002). Improved Optimization for the Robust and Accurate Linear Registration and Motion Correction of Brain Images. *NeuroImage*, *17*(2), 825–841. https://doi.org/10.1006/nimg.2002.1132

Jiang, J., Dai, B., Peng, D., Zhu, C., Liu, L., & Lu, C. (2012). Neural Synchronization during Face-to-Face Communication. *Journal of Neuroscience*, *32*(45), 16064–16069. https://doi.org/10.1523/JNEUROSCI.2926-12.2012

Jolly, E., & Chang, L. J. (2021). Multivariate spatial feature selection in fMRI. *Social Cognitive and Affective Neuroscience*, *16*(8), 795–806. https://doi.org/10.1093/scan/nsab010

Kardas, M., Schroeder, J., & O'Brien, E. (2022). Keep talking: (Mis)understanding the hedonic trajectory of conversation. *Journal of Personality and Social Psychology*, *123*(4), 717–740.

Kumar, M., Anderson, M. J., Antony, J. W., Baldassano, C., Brooks, P. P., Cai, M. B., Chen, P.-H. C., Ellis, C. T., Henselman-Petrusek, G., Huberdeau, D., Hutchinson, J. B., Li, Y. P., Lu, Q., Manning, J. R., Mennen, A. C., Nastase, S. A., Richard, H., Schapiro, A. C., Schuck, N. W., … Norman, K. A. (2021). BrainIAK: The Brain Imaging Analysis

Kit. *Aperture Neuro*, *1*(4). https://doi.org/10.52294/31bb5b68-2184-411b-8c00-a1dacb61e1da

Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest Package: Tests in Linear Mixed Effects Models. *Journal of Statistical Software*, *82*, 1–26. https://doi.org/10.18637/jss.v082.i13

Liu, L., Zhang, Y., Zhou, Q., Garrett, D. D., Lu, C., Chen, A., Qiu, J., & Ding, G. (2020). Auditory–Articulatory Neural Alignment between Listener and Speaker during Verbal Communication. *Cerebral Cortex*, *30*(3), 942–951. https://doi.org/10.1093/cercor/bhz138

Misaki, M., Kerr, K. L., Ratliff, E. L., Cosgrove, K. T., Simmons, W. K., Morris, A. S., & Bodurka, J. (2020). Beyond synchrony: The capacity of fMRI hyperscanning for the study of human social interaction. *Social Cognitive and Affective Neuroscience*, *16*(1–2), 84–92. https://doi.org/10.1093/scan/nsaa143

Nguyen, M., Chang, A., Micciche, E., Meshulam, M., Nastase, S. A., & Hasson, U. (2022). Teacher-student neural coupling during teaching and learning. *Social Cognitive and Affective Neuroscience*, *17*(4), 367–376.

Nummenmaa, L., Saarimäki, H., Glerean, E., Gotsopoulos, A., Jääskeläinen, I. P., Hari, R., & Sams, M. (2014). Emotional speech synchronizes brains across listeners and engages large-scale dynamic brain networks. *NeuroImage*, *102*, 498–509. https://doi.org/10.1016/j.neuroimage.2014.07.063

Palumbo, R. V., Marraccini, M. E., Weyandt, L. L., Wilder-Smith, O., McGee, H. A., Liu, S., & Goodwin, M. S. (2017). Interpersonal Autonomic Physiology: A Systematic Review of the Literature. *Personality and Social Psychology Review*, *21*(2), 99–141.

Pérez, P., Madsen, J., Banellis, L., Türker, B., Raimondo, F., Perlbarg, V., Valente, M., Niérat, M.-C., Puybasset, L., Naccache, L., Similowski, T., Cruse, D., Parra, L. C., & Sitt, J. D. (2021). Conscious processing of narrative stimuli synchronizes heart rate between individuals. *Cell Reports*, *36*(11), 109692.

Pickering, M. J., & Garrod, S. (2006). Alignment as the basis for successful communication. *Research on Language and Computation*, *4*(2–3), 203–228.

Prochazkova, E., Sjak-Shie, E., Behrens, F., Lindh, D., & Kret, M. E. (2021). Physiological synchrony is associated with attraction in a blind date setting. *Nature Human Behaviour*, *6*(2), 269–278.

Reed, R. G., Randall, A. K., Post, J. H., & Butler, E. A. (2013). Partner influence and in-phase versus anti-phase physiological linkage in romantic couples. *International Journal of Psychophysiology*, *88*(3), 309–316.

Regev, M., Simony, E., Lee, K., Tan, K. M., Chen, J., & Hasson, U. (2019). Propagation of Information Along the Cortical Hierarchy as a Function of Attention While Reading and Listening to Stories. *Cerebral Cortex*, *29*(10), 4017–4034. https://doi.org/10.1093/cercor/bhy282

Richard, H., Martin, L., Pinho, A. L., Pillow, J., & Thirion, B. (2019). *Fast shared response model for fMRI data* (No. arXiv:1909.12537). arXiv. https://doi.org/10.48550/arXiv.1909.12537

Schaefer, A., Kong, R., Gordon, E. M., Laumann, T. O., Zuo, X.-N., Holmes, A. J., Eickhoff, S. B., & Yeo, B. T. T. (2018). Local-Global Parcellation of the Human Cerebral Cortex from Intrinsic Functional Connectivity MRI. *Cerebral Cortex*, *28*(9), 3095–3114. https://doi.org/10.1093/cercor/bhx179

Schoot, L., Hagoort, P., & Segaert, K. (2016). What can we learn from a two-brain approach to verbal interaction? *Neuroscience & Biobehavioral Reviews*, *68*, 454–459. https://doi.org/10.1016/j.neubiorev.2016.06.009

Schwartz, L., Levy, J., Hayut, O., Netzer, O., Endevelt-Shapira, Y., & Feldman, R. (2024). Generation WhatsApp: Inter-brain synchrony during face-to-face and texting communication. *Scientific Reports*, *14*(1), 2672. https://doi.org/10.1038/s41598-024-52587-2

Sened, H., Speer, S. P. H., Cooney, G., Reece, A. G. & Tamir, D. I. The temporal arc of successful conversations. Soc. Personal. Soc. Psychol. 2024 Conference San Diego CA (2024).

Sievers, B., Welker, C., Hasson, U., Kleinbaum, A. M., & Wheatley, T. (2024). Consensus-building conversation leads to neural alignment. *Nature Communications*, *15*(1), 3936. https://doi.org/10.1038/s41467-023-43253-8

Silbert, L. J., Honey, C. J., Simony, E., Poeppel, D., & Hasson, U. (2014). Coupled neural systems underlie the production and comprehension of naturalistic narrative speech. *Proceedings of the National Academy of Sciences*, *111*(43), E4687–E4696. https://doi.org/10.1073/pnas.1323812111

Speer, S. P. H., Mwilambwe-Tshilobo, L., Tsoi, L., Burns, S. M., Falk, E. B., & Tamir, D. I. (2024a). Hyperscanning shows friends explore and strangers converge in conversation. *Nature Communications*, *15*(1), 7781. https://doi.org/10.1038/s41467-024-51990-7

Speer, S. P. H., Sened, H., Mwilambwe-Tshilobo, L., Falk, E. B., Tsoi, L., Burns, S. M., & Tamir, D. I. (2024b). Finding Agreement: fMRI-hyperscanning reveals that dyads explore in mental state space to align opinions. *bioRxiv*, 2024.10.03.616501. https://doi.org/10.1101/2024.10.03.616501

Stephens, G. J., Silbert, L. J., & Hasson, U. (2010). Speaker–listener neural coupling underlies successful communication. *Proceedings of the National Academy of Sciences*, *107*(32), 14425–14430. https://doi.org/10.1073/pnas.1008662107

Stolk, A., Noordzij, M. L., Verhagen, L., Volman, I., Schoffelen, J.-M., Oostenveld, R., Hagoort, P., & Toni, I. (2014). Cerebral coherence between communicators marks the emergence of meaning. *Proceedings of the National Academy of Sciences*, *111*(51), 18183–18188. https://doi.org/10.1073/pnas.1414886111

Stolk, A., Verhagen, L., & Toni, I. (2016). Conceptual Alignment: How Brains Achieve Mutual Understanding. *Trends in Cognitive Sciences*, *20*(3), 180–191.

Templeton, E. M., Chang, L. J., Reynolds, E. A., Cone LeBeaumont, M. D., & Wheatley, T. (2022). Fast response times signal social connection in conversation. *Proceedings of the National Academy of Sciences*, *119*(4), e2116915119. https://doi.org/10.1073/pnas.2116915119

Tsoi, L., Burns, S. M., Falk, E. B., & Tamir, D. I. (2022). The promises and pitfalls of functional magnetic resonance imaging hyperscanning for social interaction research. *Social and Personality Psychology Compass*, *16*(10), e12707. https://doi.org/10.1111/spc3.12707

Welker, C., Wheatley, T., Cason, G., Gorman, C., & Meyer, M. (2024). Self-views converge during enjoyable conversations. *Proceedings of the National Academy of Sciences*, *121*(43), e2321652121. https://doi.org/10.1073/pnas.2321652121

Wohltjen, S., & Wheatley, T. (2024). Interpersonal eye-tracking reveals the dynamics of interacting minds. *Frontiers in Human Neuroscience*, *18*, 1356680.

Yeomans, M., Schweitzer, M. E., & Brooks, A. W. (2022). The Conversational Circumplex: Identifying, prioritizing, and pursuing informational and relational motives in conversation. *Current Opinion in Psychology*, *44*, 293–302.

Yeshurun, Y., Swanson, S., Simony, E., Chen, J., Lazaridi, C., Honey, C. J., & Hasson, U. (2017). Same Story, Different Story: The Neural Representation of Interpretive Frameworks. *Psychological Science*, *28*(3), 307–319. https://doi.org/10.1177/0956797616682029

Zada, Z., Goldstein, A., Michelmann, S., Simony, E., Price, A., Hasenfratz, L., Barham, E., Zadbood, A., Doyle, W., Friedman, D., Dugan, P., Melloni, L., Devore, S., Flinker, A., Devinsky, O., Nastase, S. A., & Hasson, U. (2024). A shared model-based linguistic space for transmitting our thoughts from brain to brain in natural conversations. *Neuron*, *112*(18), 3211-3222.e5.

Zada, Z., Nastase, S. A., Speer, S., Mwilambwe-Tshilobo, L., Tsoi, L., Burns, S., Falk, E., Hasson, U., & Tamir, D. (2025). Linguistic coupling between neural systems for

speech production and comprehension during real-time dyadic conversations. *bioRxiv*, 2025.02.14.638276. https://doi.org/10.1101/2025.02.14.638276

Zadbood, A., Chen, J., Leong, Y. C., Norman, K. A., & Hasson, U. (2017). How We Transmit Memories to Other Brains: Constructing Shared Neural Representations Via Communication. *Cerebral Cortex (New York, NY)*, *27*(10), 4988–5000. https://doi.org/10.1093/cercor/bhx202